MILITARY SPECIFICATION

INTEGRATED CIRCUIT, DIGITAL, CMOS LOGIC ARRAY MONOLITHIC SILICON

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, test, manufacturing and acceptance requirements for the logic array integrated circuit (I.C.) to be used in the M718E1/M741E1 Mines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extend specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510	-	Microcircuits, general specification for.
MIL-I-45208	-	Inspection system requirements.
MIL-M-55565	-	Microcircuits, preparation for delivery of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling procedures and Tables for inspection by attributes.
MIL-STD-883	-	Test methods and procedures for microelectronics.
MIL-STD-1331	-	Parameters to be controlled for the specification of micro-circuits.
MIL-STD-130	-	Identification marking of U.S. military property.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extend herein.

DRAWINGS

U. S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

PRODUCT AND PACKAGING DRAWINGS

9342578	-	Logic Diagram, Logic Array, CMOS
9317646	-	Qualification, Electronic Component.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Parts and Materials.

3.1.1 Components. The I.C. shall comply with all the requirements of the applicable drawings, specifications and standards.

3.1.2 Metals. External metal surface shall be corrosion resistant or shall be plated or treated to resist corrosion. External leads shall meet the requirements specified in 3.5.3.

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3.1.3 Other materials. External parts, elements or coatings including markings shall not blister, crack, soften, flow or exhibit defect that adversely affect storage, operation or environmental capabilities of the integrated circuit delivered to this specification under the specified test condition.

3.2 Design and construction. I.C. design and construction shall be in accordance with the requirements specified herein and in the applicable specifications or drawings.

3.2.1 Dimensions. The package dimension shall conform to MIL-M-38510, Appendix C, Case Outline D6, Configuration 1.

3.2.2 Package. The finished component shall be supplied in a 18 pin, dual-in-line hermetically sealed package. The package shall be of the type shown in Figure 1, utilizing a ceramic lid, a ceramic base with gold deposit for mounting the die, Alloy 42 or Kovar solid metal lead frame thru to cavity with aluminum cladding, and a glass frit to accomplish the seal between the lid, lead frame and the base frit seal temperature shall be in excess of 400°C. No organic or polymeric material shall be used inside the package.

3.2.3 Passivation. The chip shall be passivated with silicon dioxide and/or silicon nitrade to render it impervious to contaminated environments. The bonding pad areas shall be the only areas with no surface passivation.

3.2.4 Static charge protection. Static charge protection shall be provided internally on all terminals.

3.2.5 Pin assignments shall be as shown in Figure 2.

3.3 Design documentation. When specified in the procurement document, design, topography and schematic circuit information for the microcircuit supplied under this specification shall be submitted to the procuring activity. Unless otherwise specified, all design documentation shall be sufficient to depict completely the physical and electrical construction of the microcircuit supplied under this specification, and shall be traceable to the specific production lot(s) and inspection lot codes under which microcircuit are manufactured and tested so that revisions can be identified.

3.3.1 Die topography. For microcircuit dies, there shall be an enlarged color photograph(s) showing the topography of elements of the die to a minimum magnification of 80X. If this results in a photograph larger than 8" x 10", the magnification may be reduced to accommodate an 8" x 10" view.

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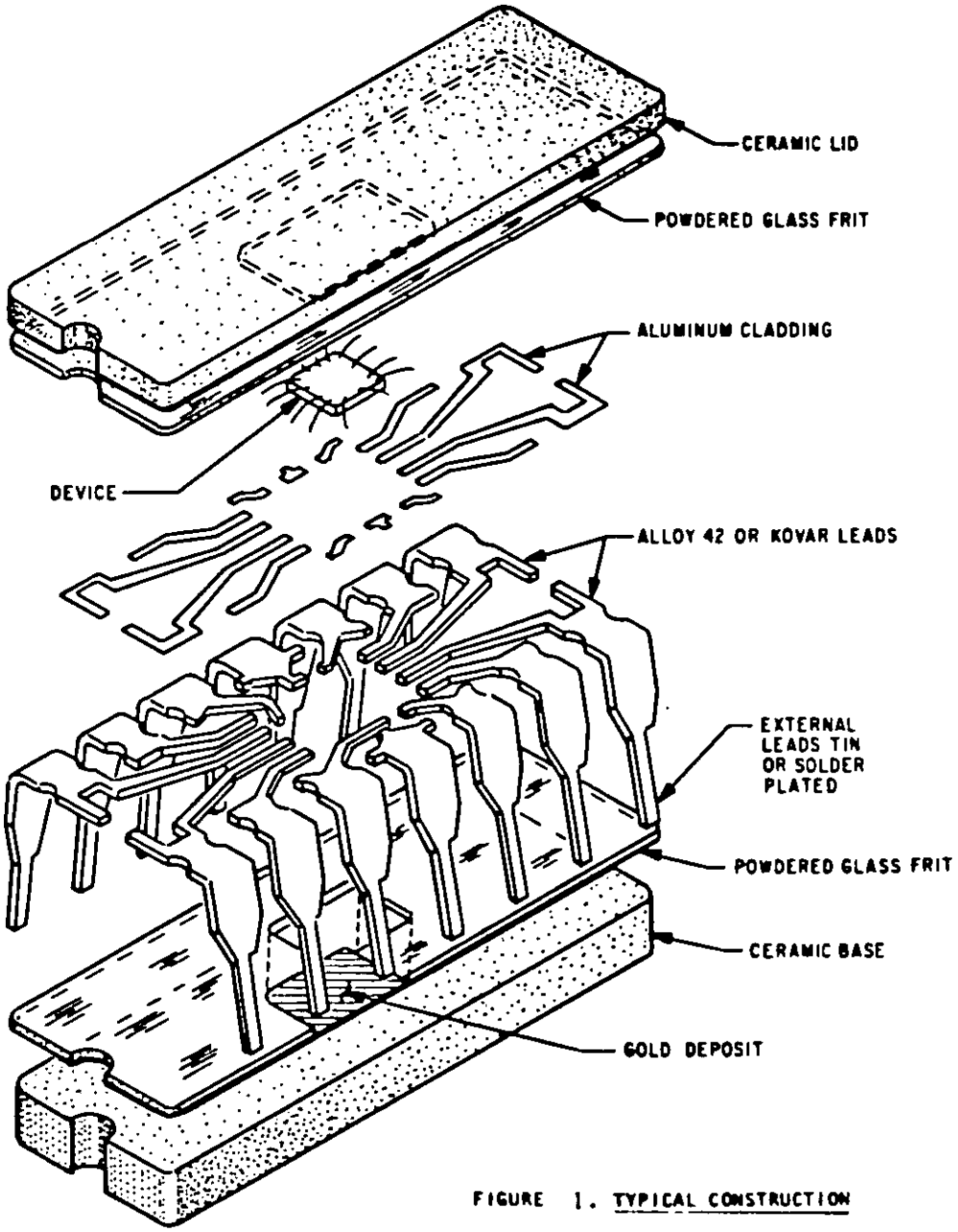


FIGURE 1. TYPICAL CONSTRUCTION

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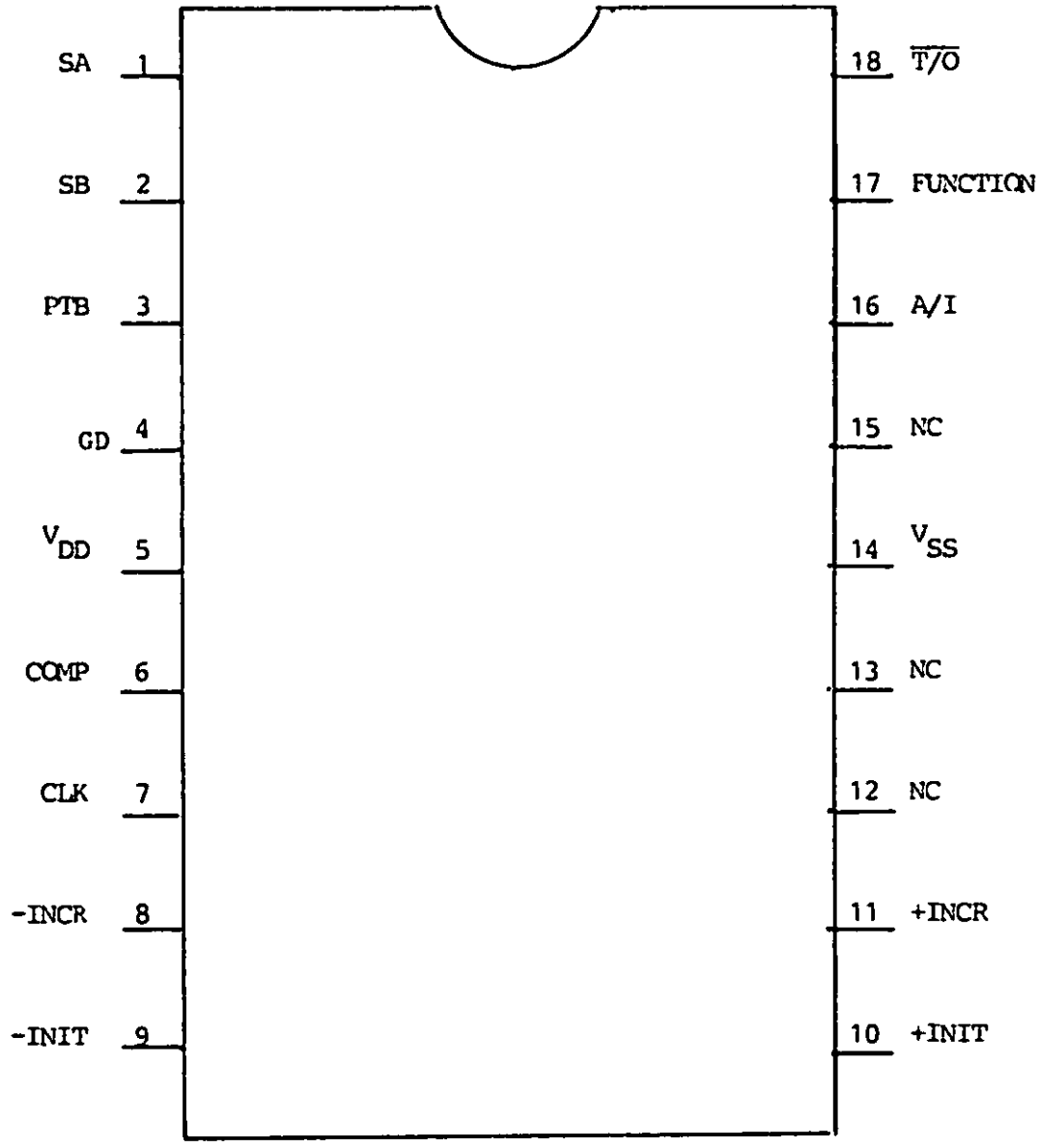


FIGURE 2 PIN ASSIGNMENT

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3.3.2 Logic diagram. The actual logic diagrams of the microcircuit supplied under this specification shall be provided, showing all the logic/circuit elements functionally designed into the microcircuit together with their values, when applicable.

3.3.3 Country of manufacturer. The component specified herein shall be manufactured, assembled and tested within the USA or its territories or by a foreign source complying with the provisions of the defense standardization manual, DOD 4120.3-M, paragraph 4-106.6, subject to the approval of the Government Technical Agency. (See 6.4).

3.4 Electrical requirements.

3.4.1 Maximum Rating. (As specified in Table 1)

TABLE 1. MAXIMUM RATING

Characteristic	Symbol	Rating	Unit
D C Supply Voltage (See 3.4.3)	V_{DD}	-0.3 to +10	Volts
All Inputs	V_{IN}	-0.3 to +10	Volts
Temperature			
Operating	$T_A(OP)$	-55 to +125	^oC
Storage	$T_A(STG)$	-65 to +150	^oC

3.4.2 Performance characteristics. (As specified in Table 2)

3.4.2.1 The I.C. circuit shall function in accordance with the logic diagram as specified in dwg. 9342578 when operated under power supply requirements as specified in paragraph 3.4.3.

3.4.3 Power supply requirements. The component shall be capable of functioning with a power supply having the following requirements.

3.4.3.1 The output impedance of the supply source shall be 40 ohms minimum, 1,000 ohms maximum and shall be a 1.0 uf minimum capacitor across its output.

3.4.3.2 The supply voltage shall rise to 96% of its maximum value in less than 0.5 seconds. Noise voltage on the supply output shall not exceed 5% V_{OD} peak to peak and its frequency shall not exceed 100 KHz.

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3.4.3.3 Electrical tests. Electrical tests shall verify conformance to the requirements of Tables II and III of this specification. The component input to output coding shall be as described in the test pattern shown in Table III. The pattern in Table III shall be run with VDD values of 5 and 10 volts respectively. The pattern should be repeated at the 3 VDD values with all output pins open (uploaded) to meet the dynamic IDD requirements of Table II. In addition, Pin 3 (PTB) should provide a divide by 384 function (16 2/3% duty cycle) of the input at Pin 7. This output to be checked at any time during the test.

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TABLE II. PERFORMANCE CHARACTERISTICS: T= 25°C

PARAMETER	TEST CONDITIONS	OUTPUT PINS NO.	OUTPUT CLOCK CYCLE	OUTPUT VOLTAGE (VOLTS) MAX	MIN
V_{OH}	V_{DD} = 5.0V I_{OH} = 0.1ma Connect a 46.4K resistor to V_{SS}	1.2.9.17 10 11 8 3	88466 0 2051 2052 Note 1	4.6	N/A
	V_{DD} = 10.0V I_{OH} = 0.2ma Connect a 95.3K resistor to V_{SS}	1,2,9,17 10 11 8 3	88466 0 2051 2052 Note 1	9.5	N/A
V_{OL}	V_{DD} = 5.0V I_{OH} = 0.1ma Connect a 23.2K resistor to V_{DD}	8,9,11 10 3	0 88466 Note 1		0.40
	V_{OD} = 10.0V I_{OH} = 0.2ma Connect a 47.5K reistor to V_{DD}	8,9,11 10 3	0 88466 Note 1		0.50
	V_{DD} = 5.0V I_{OH} = 0.1ma Connect a 11.5K resistor to V_{DD}	1,2,17	0		0.40
	V_{DD} = 5.0V I_{OH} = 0.1 ma Connect a 23.7K resistor to V_{DD}	1,2,17	0		0.40

Note 1: This output (PTB) is not reset by T/O (Pin 18). PTB stays high for 64 clock cycles and low for 320 clock cycles and may be measured at any clock count which achieves the desired state.

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TABLE II. PERFORMANCE CHARACTERISTICS: T= 25°C

PARAMETER	PINS NO.	TEST CONDITIONS	MAX	MIN	UNITS
V_{IH}	6, 7, 16, 18	V_{DD} = 5.0V	3.5		
		V_{DD} = 10. 0V	7.0		
V_{IL}	6, 7, 16, 18	V_{DD} = 5.0V		1.5	VOLTS
		V_{DD} = 10.0V		3.0	
V_{IN}	6, 7, 16, 18	V_{IN} = 5.0V		-0.1	
		V_{IN} = 10.0V		-0.2	
V_{IL}	6, 7, 16, 18	V_{IN} = 0.2V		0.1	VOLTS
		V_{IN} = 0.5V		0.2	
I_{DD}	5 V_{DD} = 7.0V	STATIC All inputs @ V_{IL}, outputs unloaded	10		
		DYNAMIC Input Pattern of Table III @ 1 KH_Z Clock rate	100		MA
T_{PD} and other timing parameters	Compliance with paragraph 3.4 @ V_{DD} = 5.0 Volts and CLK = 1.0 KH_Z (Pin 7) shall constitute verification that internal propagation delay and timing parameters satisfy the requirements of this specification.				

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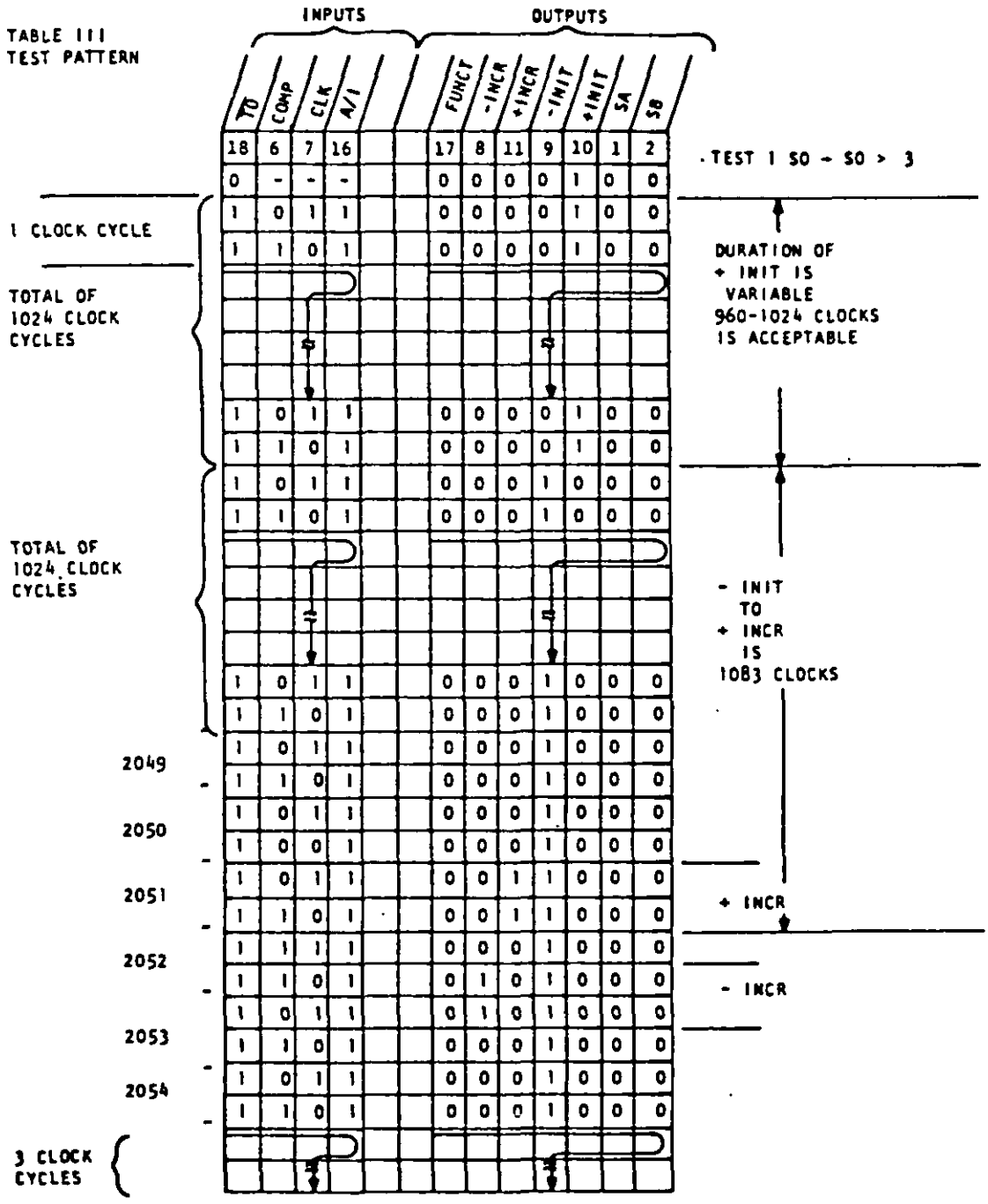


TABLE III. TEST PATTERN.

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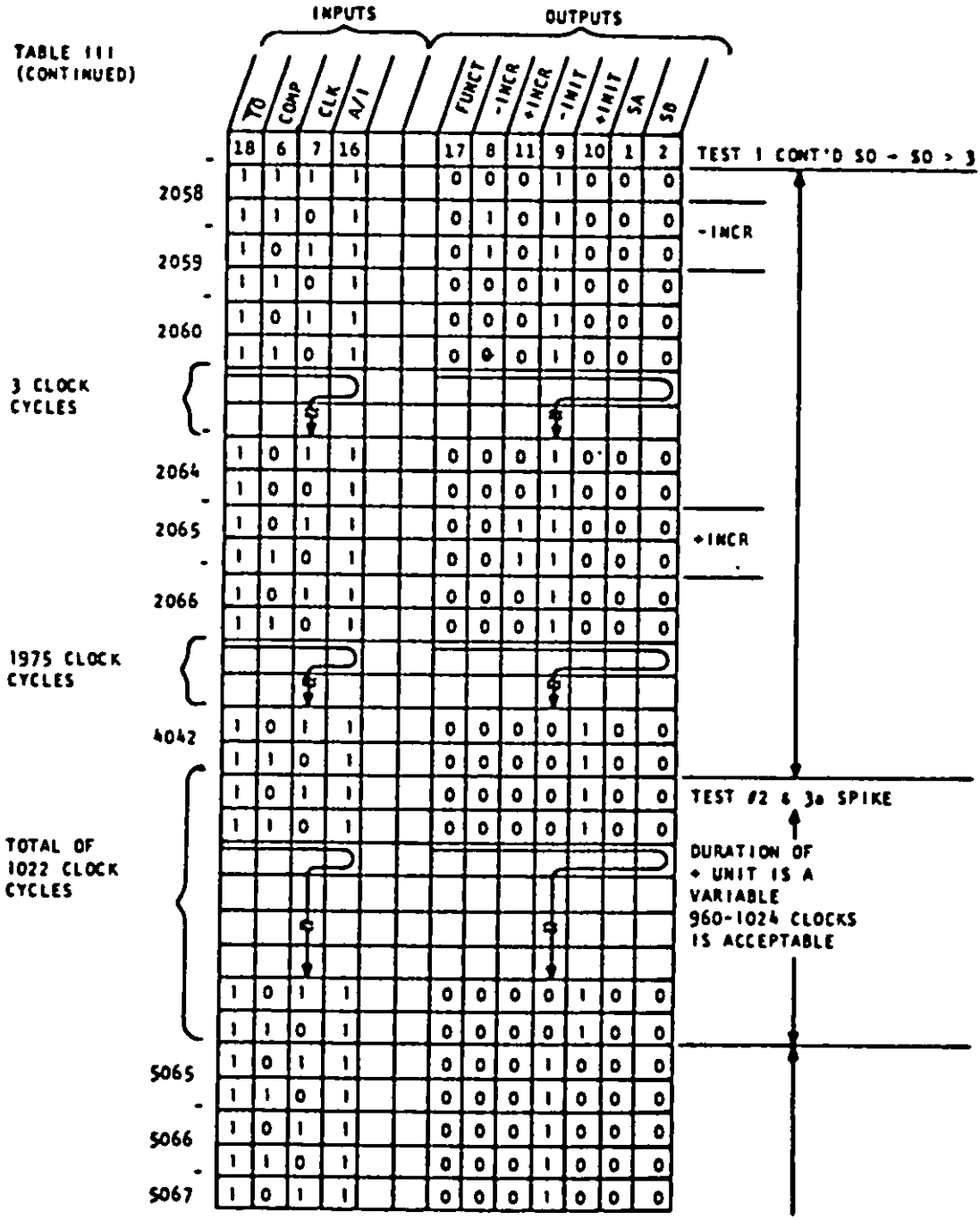


TABLE III. (Cont'd)

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TABLE III
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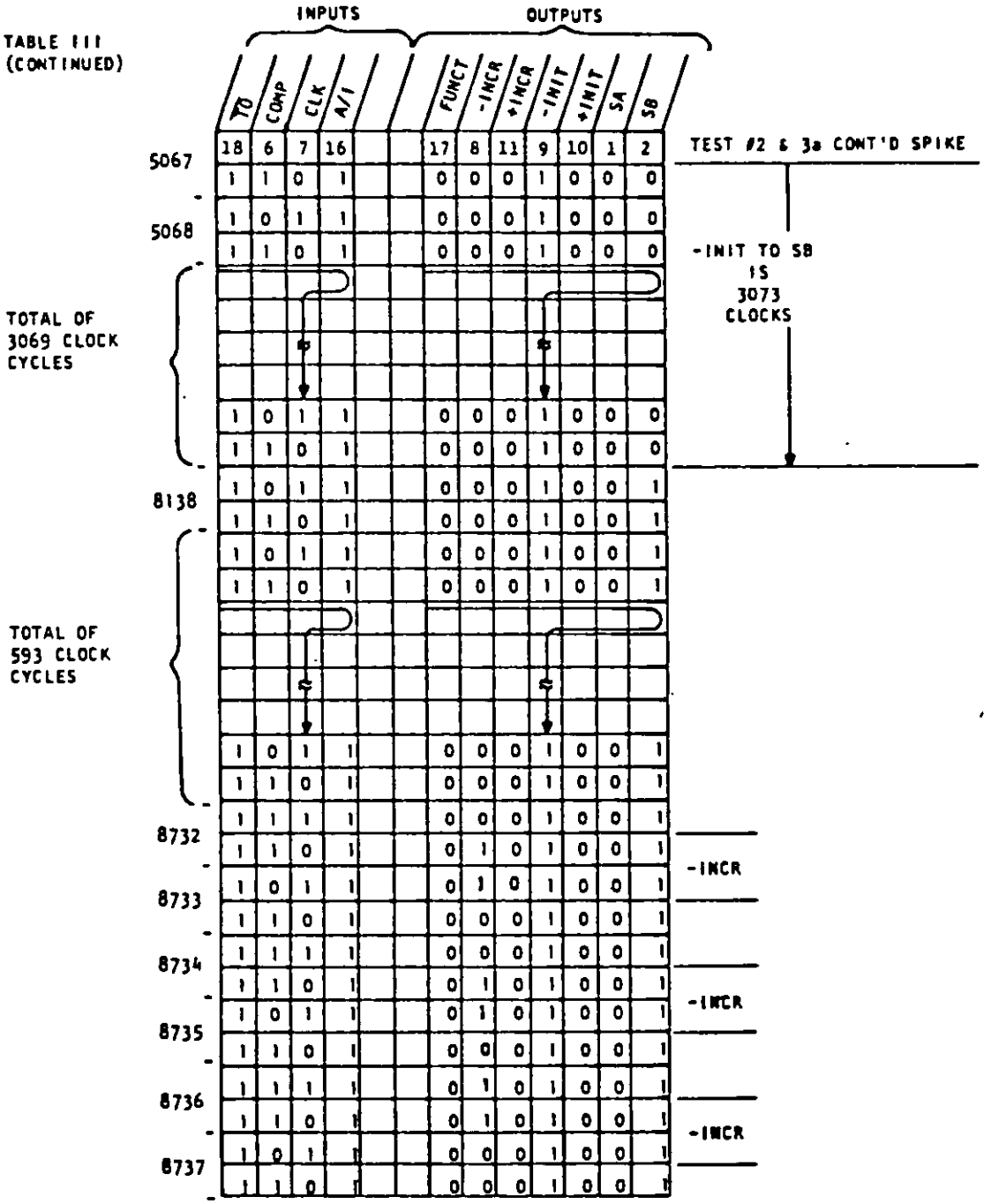


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TABLE III
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	INPUTS					OUTPUTS							
	TO	COMP	CLK	A/I		FUNCT	-INCR	+INCR	-INIT	+INIT	SA	SB	
	18	6	7	16		17	8	11	9	10	1	2	TEST #2 & 3a CONT'D SPIKE
8738	1	1	1	1		0	0	0	1	0	0	1	
	1	1	0	1		0	1	0	1	0	0	1	-INCR
8739	1	0	1	1		0	1	0	1	0	0	1	
	1	1	0	1		0	0	0	1	0	0	1	
8740	1	1	1	1		0	0	0	1	0	0	1	
	1	1	0	1		0	1	0	1	0	0	1	-INCR
8741	1	0	1	1		0	1	0	1	0	0	1	
	1	1	0	1		0	0	0	1	0	0	1	
8742	1	1	1	1		0	0	0	1	0	0	1	
	1	1	0	1		0	1	0	1	0	0	1	-INCR
8743	1	0	1	1		0	1	0	1	0	0	1	
	1	1	0	1		0	0	0	1	0	0	1	
8744	1	1	1	1		0	0	0	1	0	0	1	
	1	1	0	1		0	1	0	1	0	0	1	-INCR
8745	1	0	1	1		0	1	0	1	0	0	1	
	1	1	0	1		0	0	0	1	0	0	1	
8746	1	1	1	1		0	0	0	1	0	0	1	
	1	1	0	1		0	1	0	1	0	0	1	
8747	1	1	1	1		0	1	0	1	0	1	0	
	1	1	0	1		0	1	0	1	0	1	0	-INCR
8748	1	1	1	1		0	1	0	1	0	1	0	
	1	1	0	1		0	1	0	1	0	1	0	
8749	1	1	1	1		0	1	0	1	0	1	0	
	1	1	0	1		0	1	0	1	0	1	0	
8750	1	0	1	1		0	1	0	1	0	1	0	
	1	1	0	1		0	0	0	1	0	1	0	
8751	1	0	1	1		0	0	0	1	0	1	0	
	1	0	0	1		0	0	0	1	0	1	0	
8752	1	0	1	1		0	0	1	1	0	1	0	+INCR
	1	0	0	1		0	0	1	1	0	1	0	
8753	1	0	1	1		0	0	1	1	0	1	0	

TABLE III. (Cont'd)

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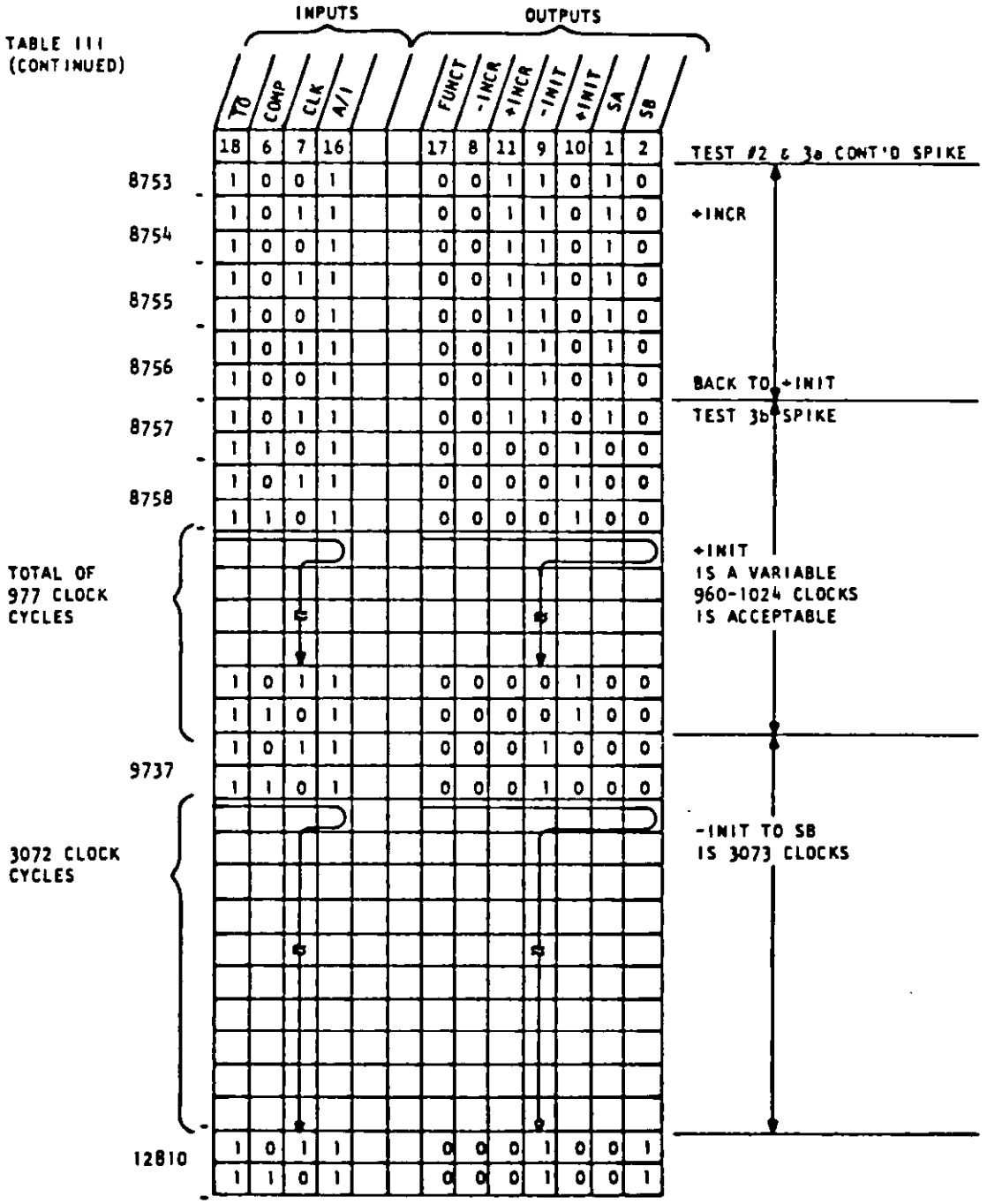


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TABLE III
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	INPUTS					OUTPUTS							
	TO	COMP	CLK	A/I		FUNCT	-INCR	+INCR	-INIT	+INIT	SA	SD	TEST 36 CONT'D SPIKE
	18	6	7	16		17	8	11	9	10	1	2	
	1	0	1	1		0	0	0	1	0	0	1	
12811	1	1	0	1		0	0	0	1	0	0	1	
51 CLOCK CYCLES													
	1	0	1	1		0	0	0	1	0	0	1	
12863	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12864	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
12865	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
12866	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
12867	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12868	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
12869	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12870	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
12871	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12872	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
12873	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12874	1	1	0	1		0	0	1	1	0	0	1	+INCR
12875	1	0	1	1		0	0	0	1	0	0	1	

TABLE III. (Cont'd)

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TABLE III
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	INPUTS					OUTPUTS							
	TO	COMP	CLK	A/I		FUNCT	-INCR	+INCR	-INIT	+INIT	SA	SB	
	18	6	7	16		17	8	11	9	10	1	2	TEST 3b CONT'D SPIKE
12875	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
12876	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
12877	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
12878	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
12879	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
12880	1	0	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	1	1	0	1	0	
12881	1	0	0	1		0	0	1	1	0	1	0	+INCR
	1	0	1	1		0	0	1	1	0	1	0	
12882	1	1	0	1		0	0	1	1	0	1	0	
	1	1	1	1		0	0	0	1	0	1	0	
12883	1	1	0	1		0	1	0	1	0	1	0	
	1	1	1	1		0	1	0	1	0	1	0	
12884	1	1	0	1		0	1	0	1	0	1	0	
	1	1	1	1		0	1	0	1	0	1	0	BACK TO +INIT
12885	1	1	0	1		0	1	0	1	0	1	0	
	1	1	1	1		0	1	0	1	0	1	0	
12886	1	1	0	1		0	1	0	1	0	1	0	
	1	1	1	1		0	1	0	1	0	1	0	
12887	1	1	0	1		0	0	0	0	1	0	0	
	1	0	1	1		0	0	0	0	1	0	0	TEST 4 t2
12888	1	1	0	1		0	0	0	0	1	0	0	DURATION OF +INIT IS A VARIABLE 960-1024 CLOCKS ARE ACCEPTABLE t2
1008 CLOCK CYCLES													
	1	0	1	1		0	0	0	1	0	0	0	
13897	1	1	0	1		0	0	0	1	0	0	0	

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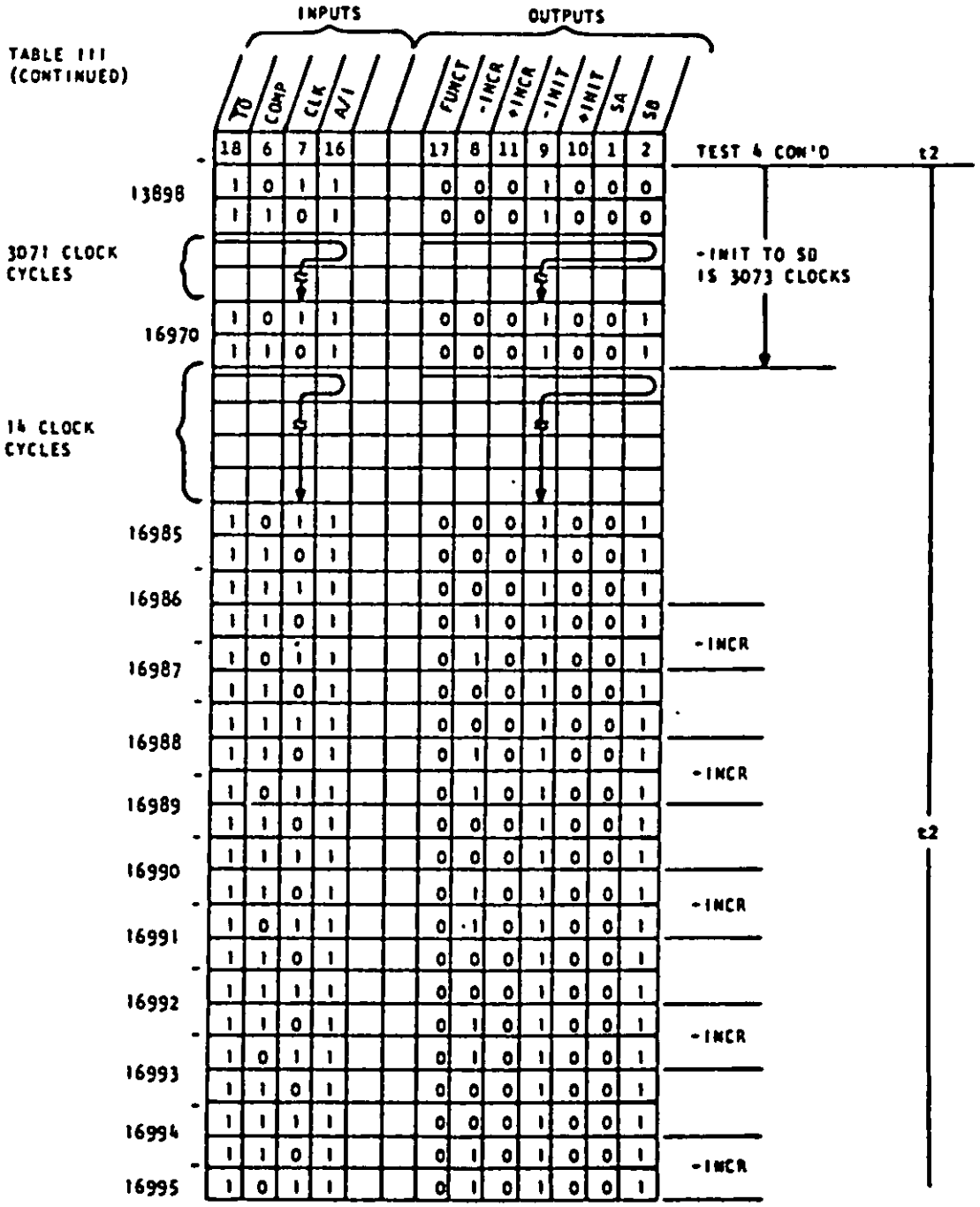


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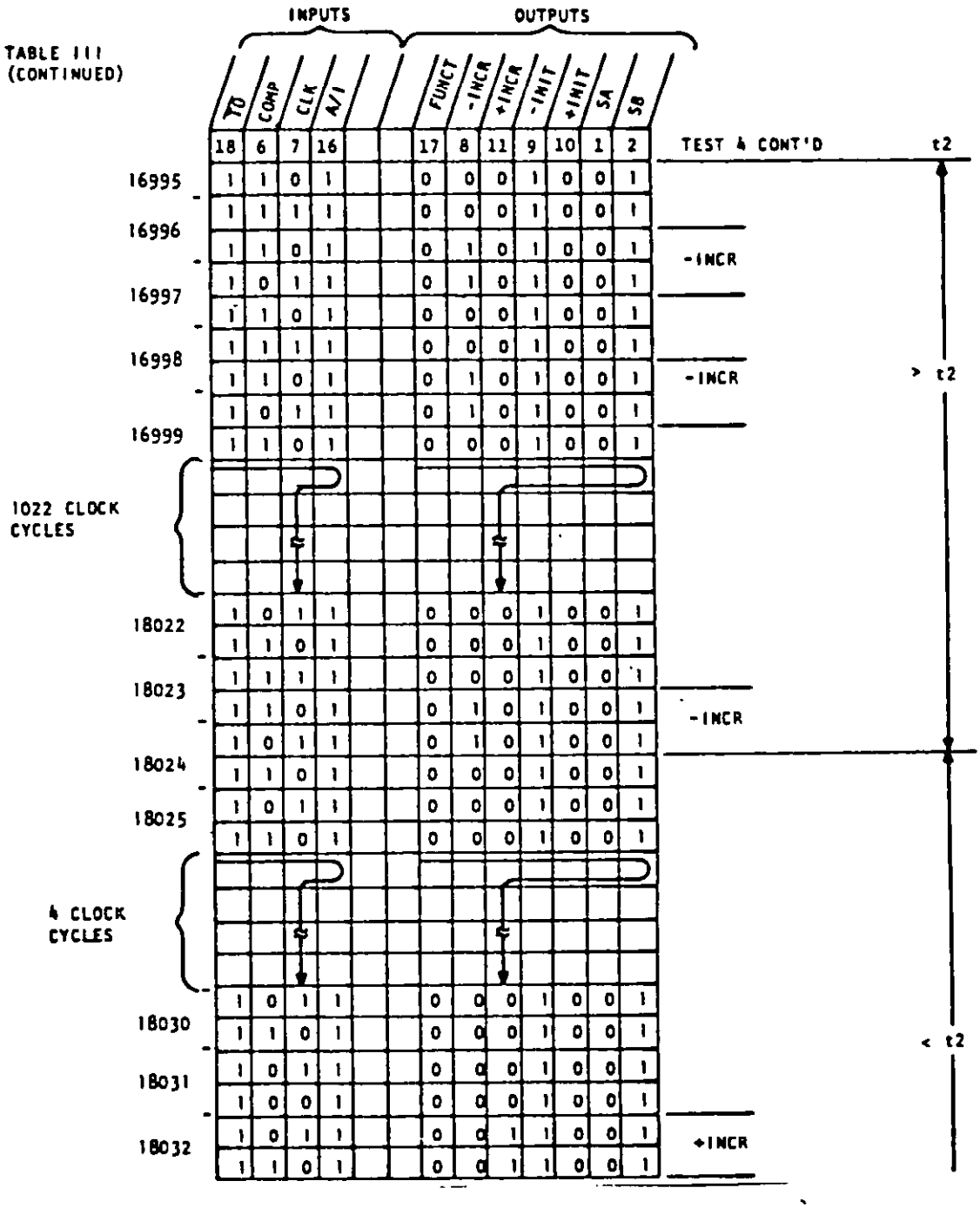


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TABLE III
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	INPUTS					OUTPUTS							
	TO	COMP	CLK	ADJ		FUNCT	-INCR	+INCR	-INIT	+INIT	SA	SB	
	18	6	7	16		17	8	11	9	10	1	2	TEST & CONT'D
	1	0	1	1		0	0	0	1	0	0	1	t2
18033	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18034	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18035	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
18036	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
18037	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
18038	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18039	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
18040	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	<t2
18041	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
18042	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18043	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
18044	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18045	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
18046	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
18047	1	0	0	1		0	0	0	1	0	0	1	
18048	1	0	1	1		0	0	1	1	0	0	1	+INCR

TABLE III. (Cont'd)

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TABLE III
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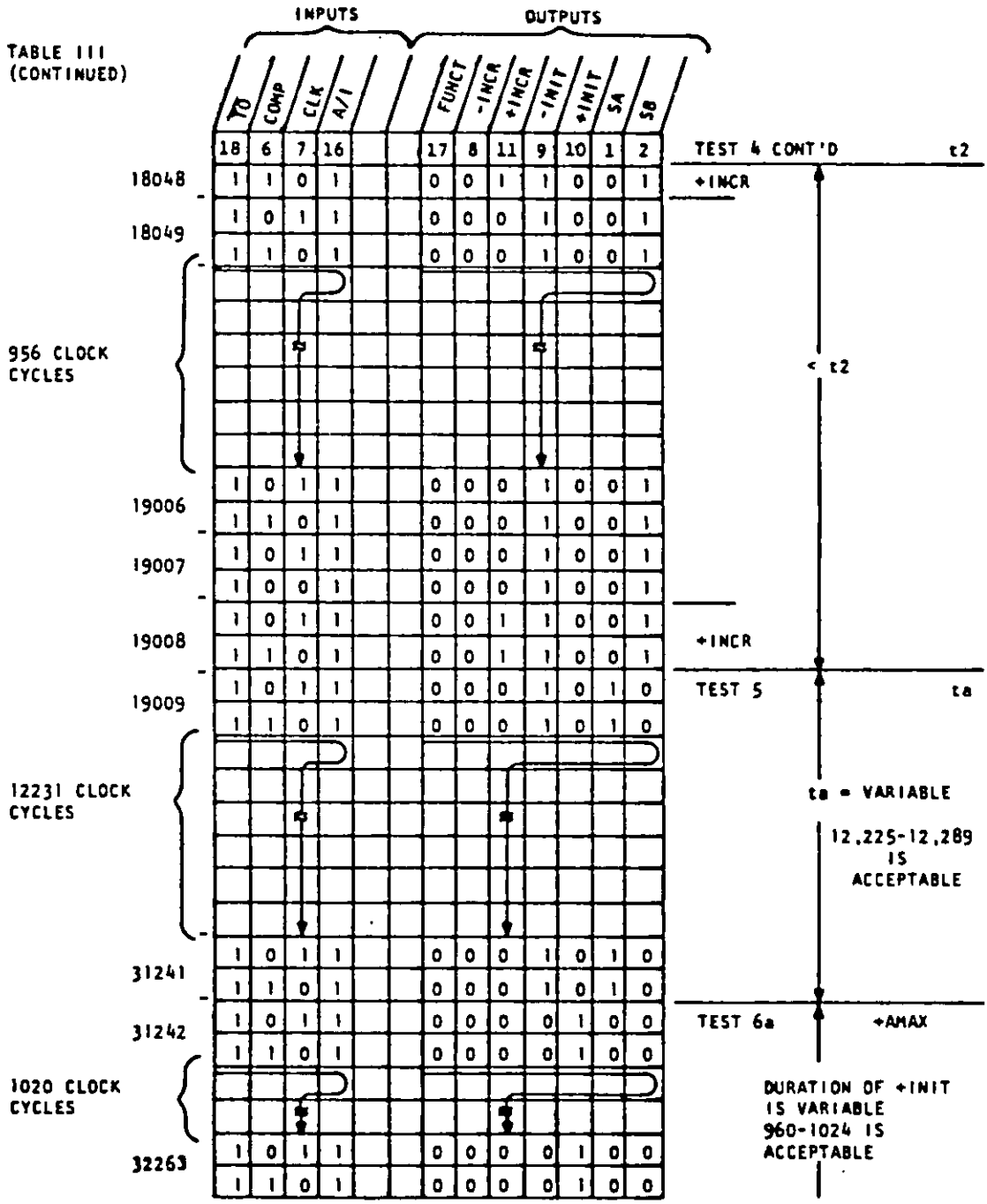


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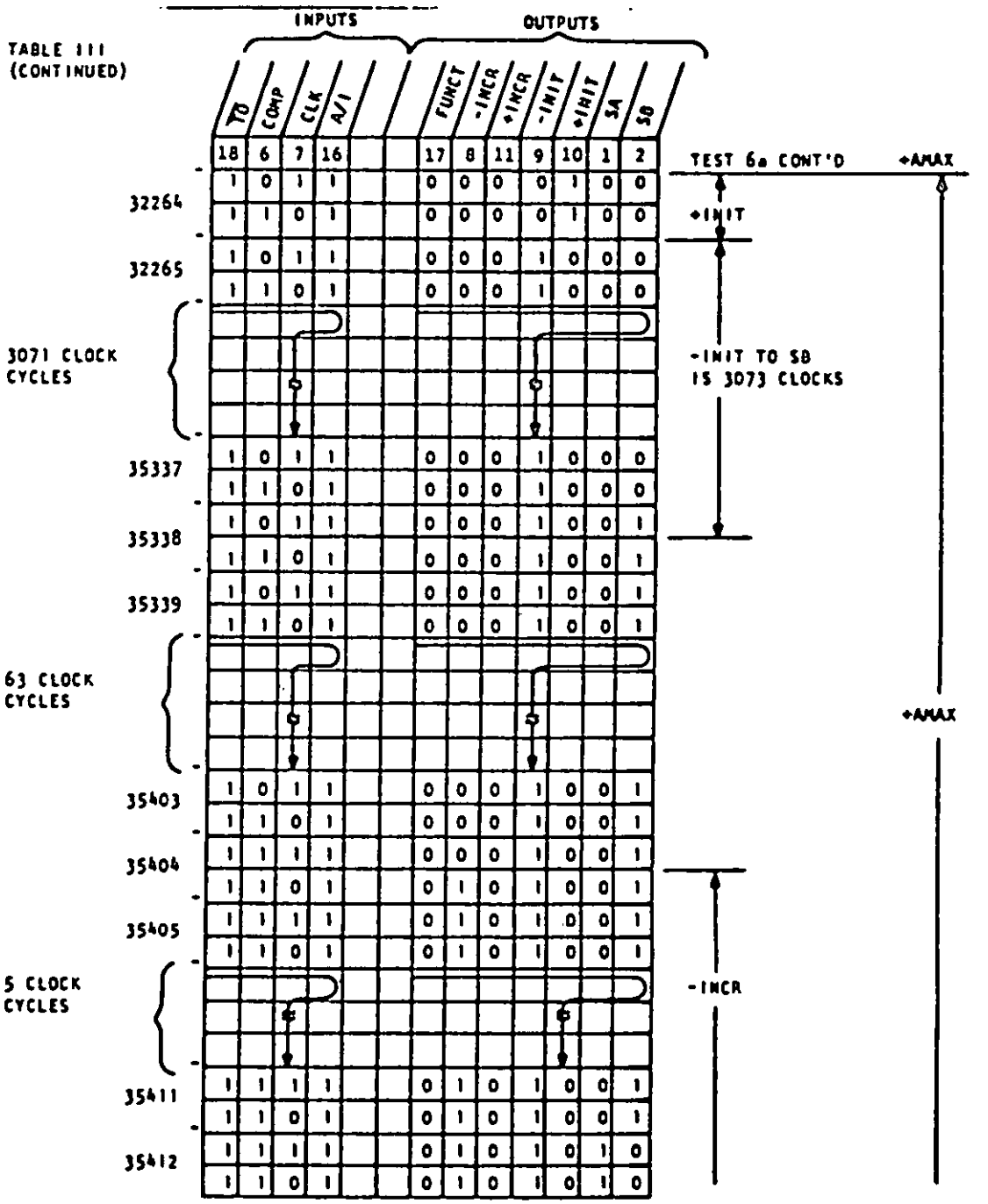


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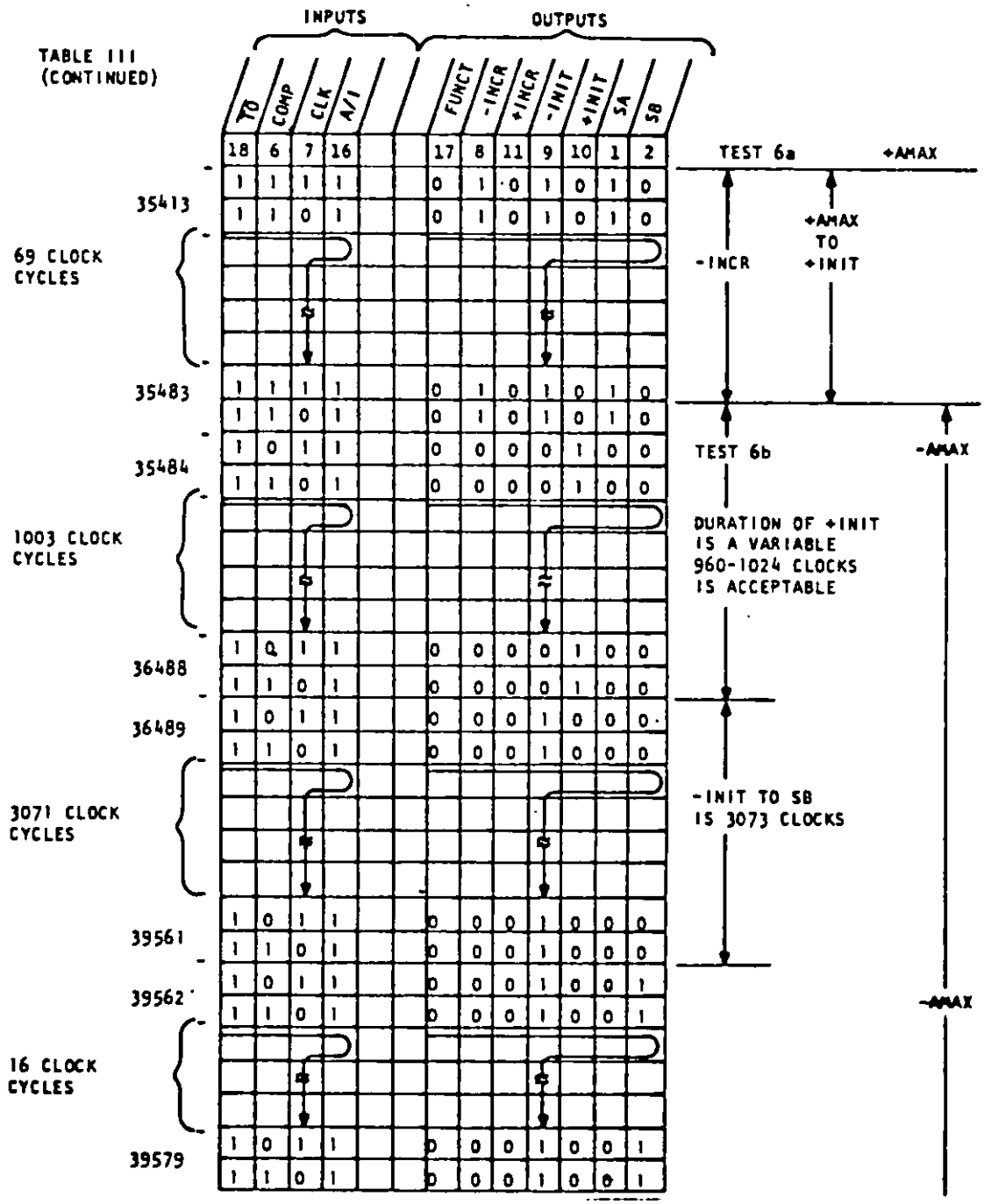


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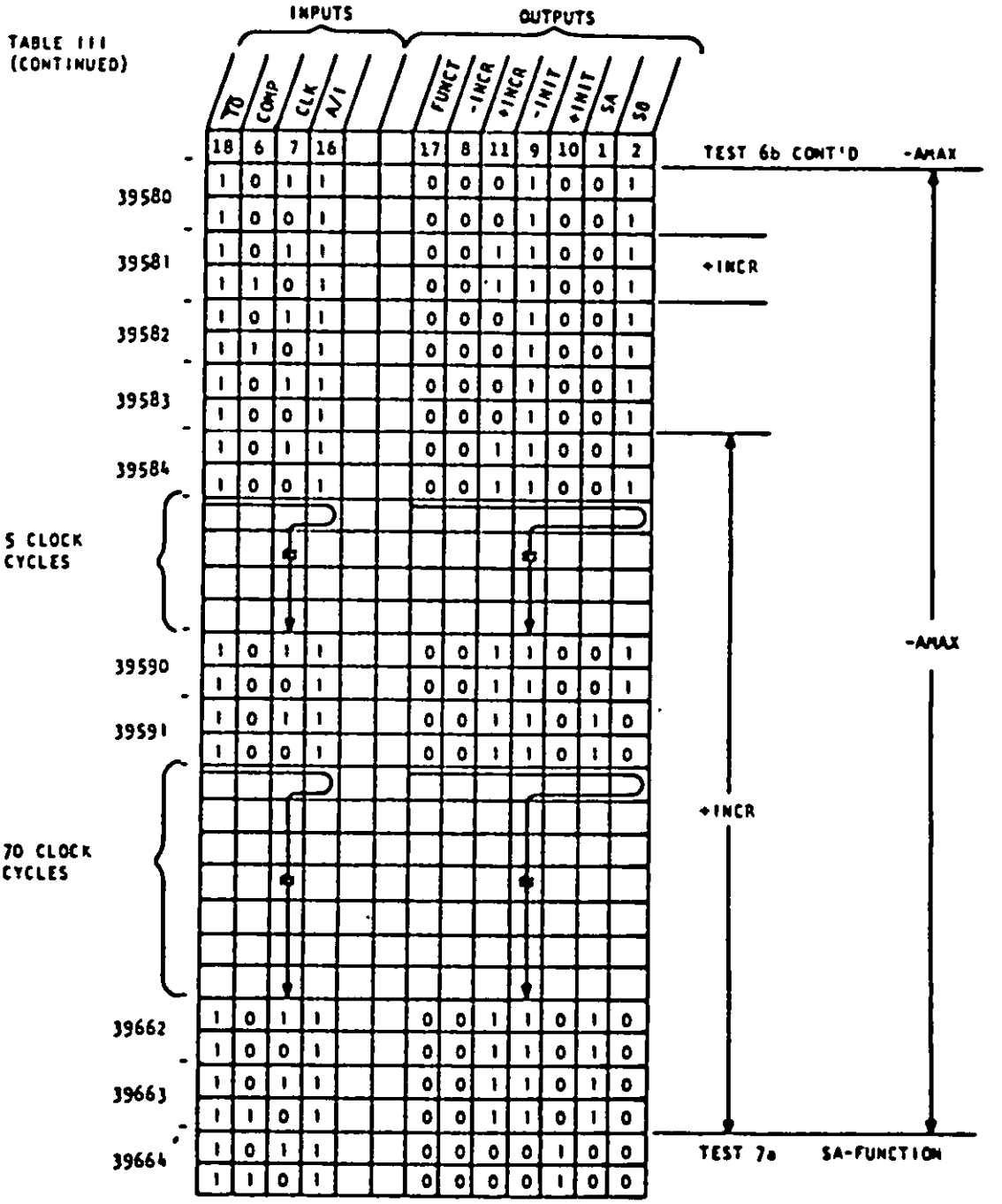


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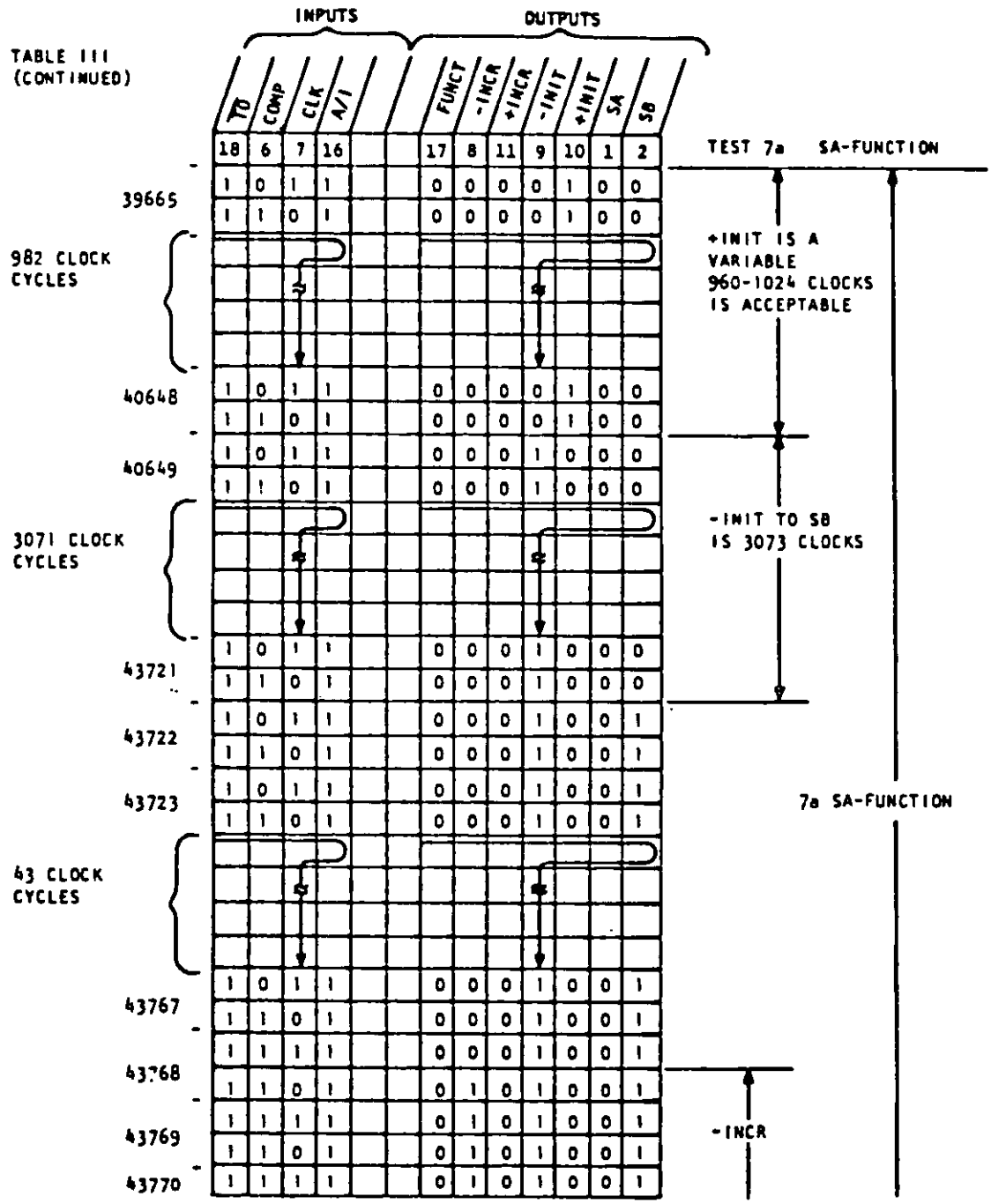


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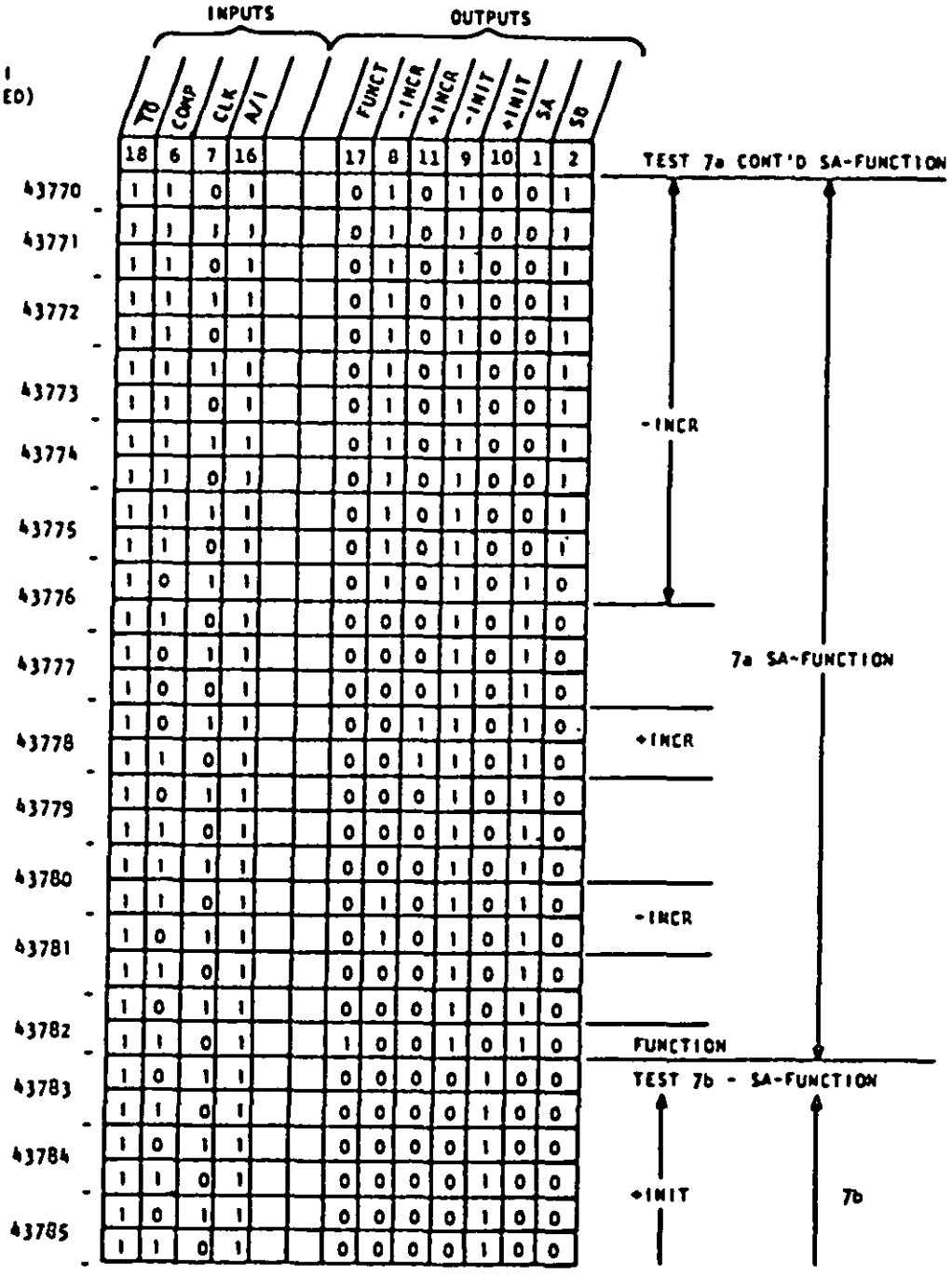


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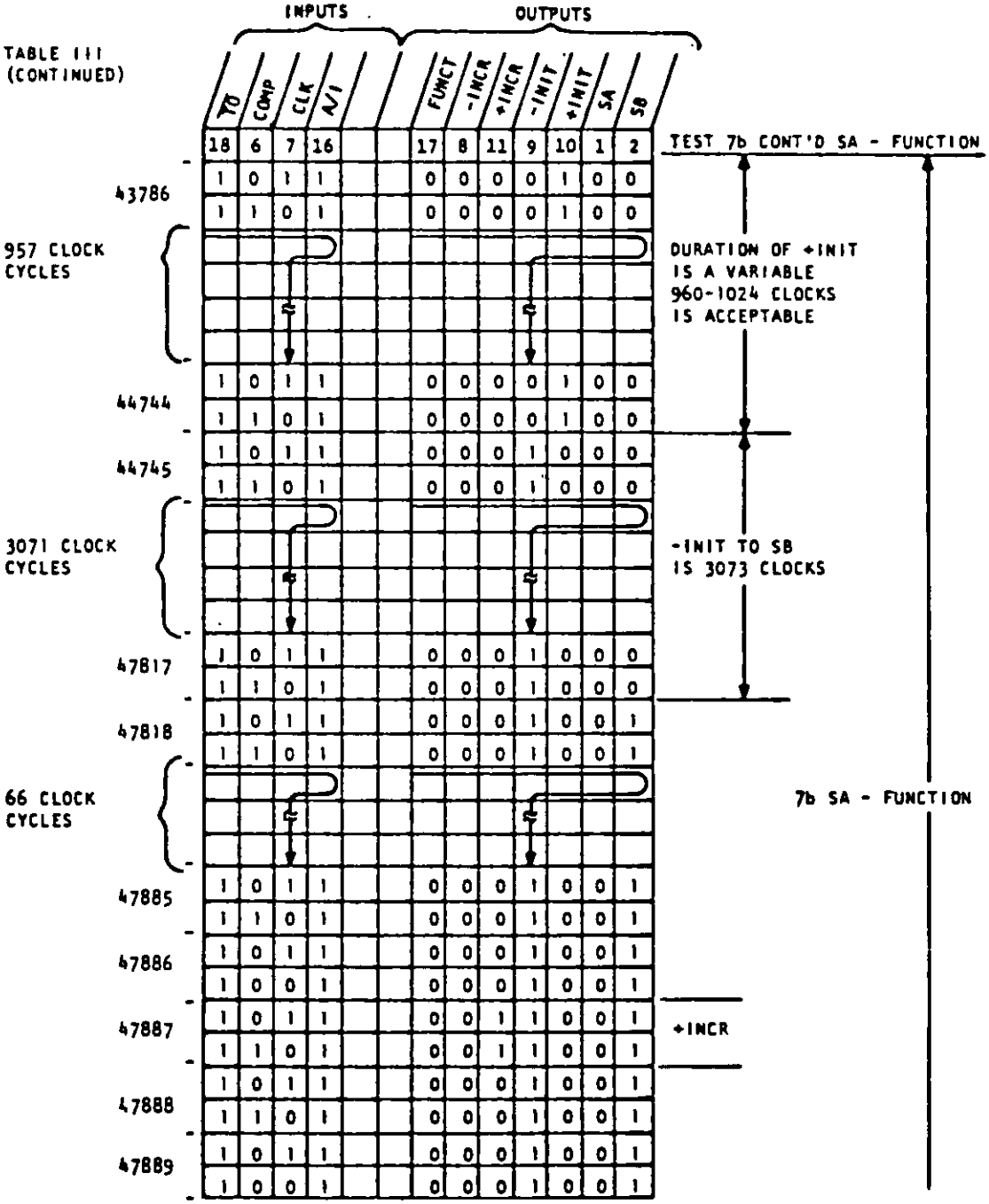


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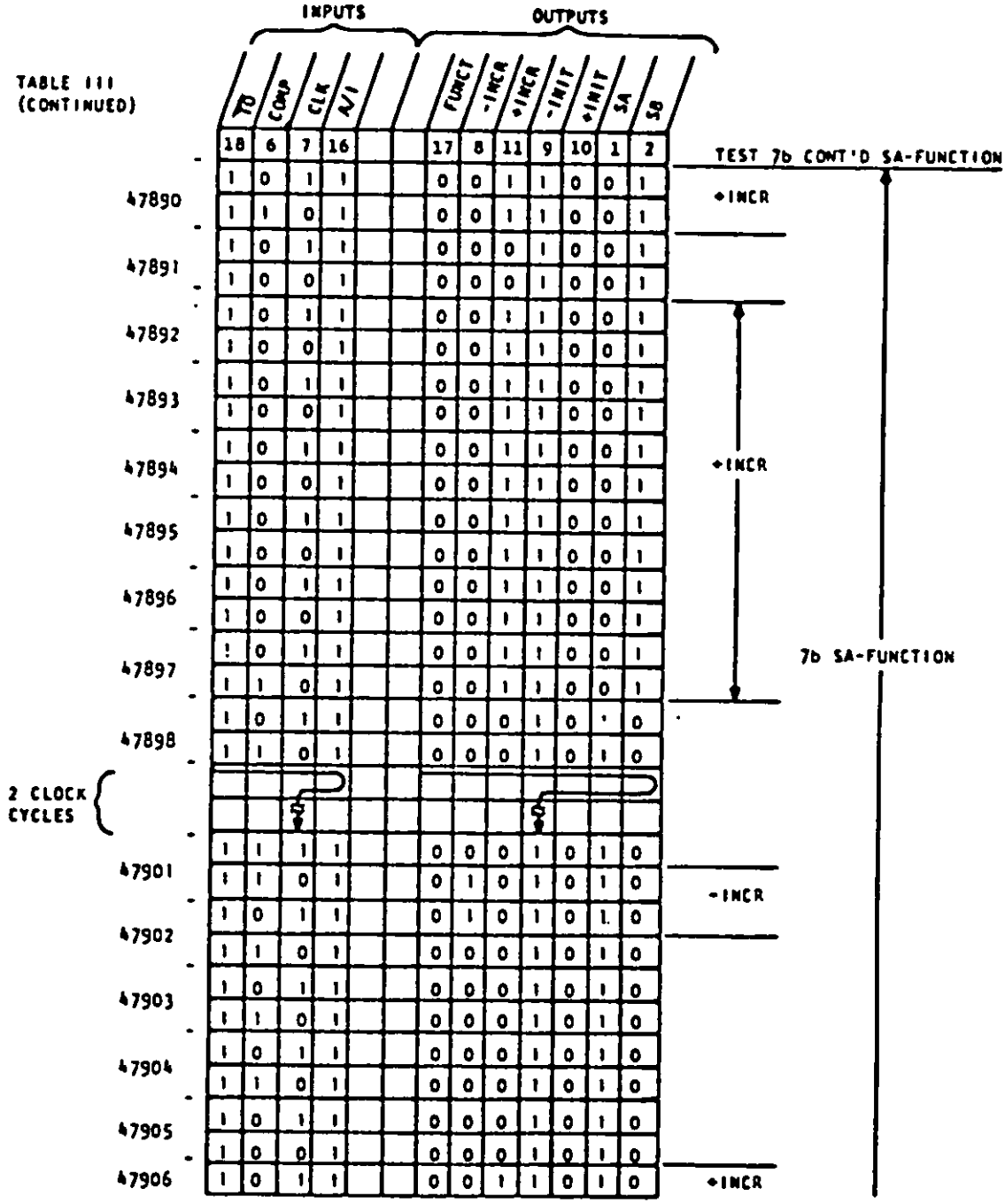


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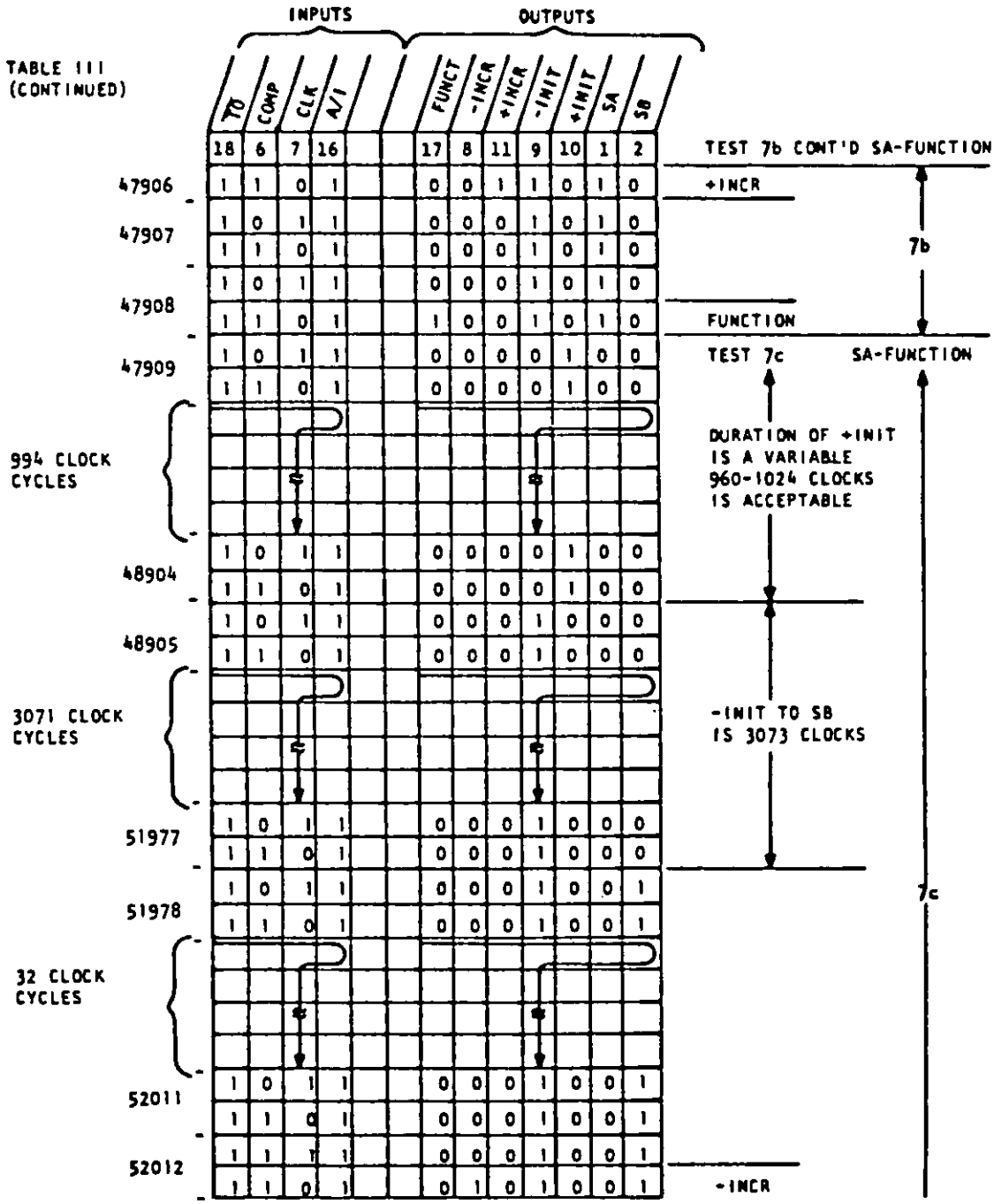


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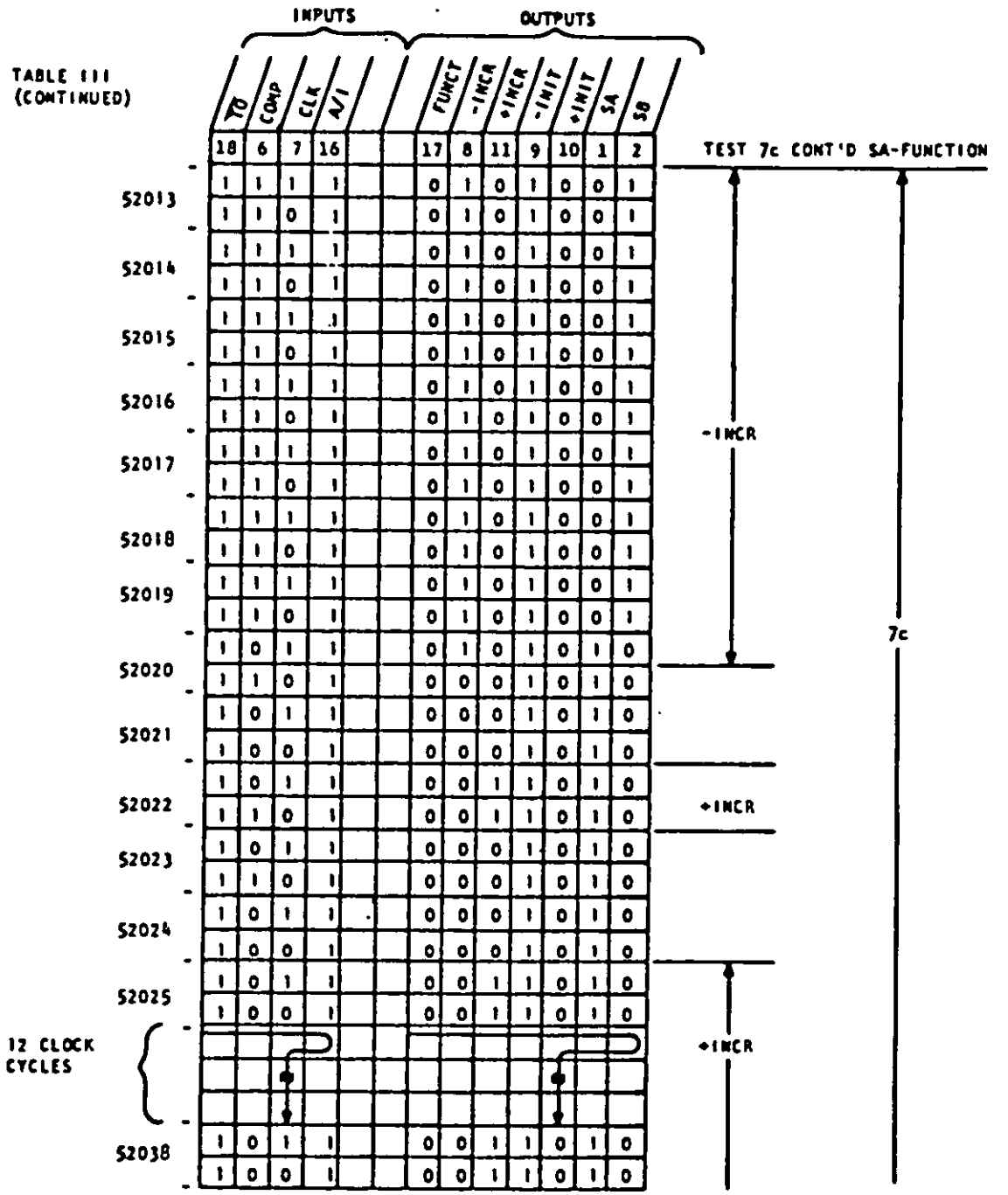


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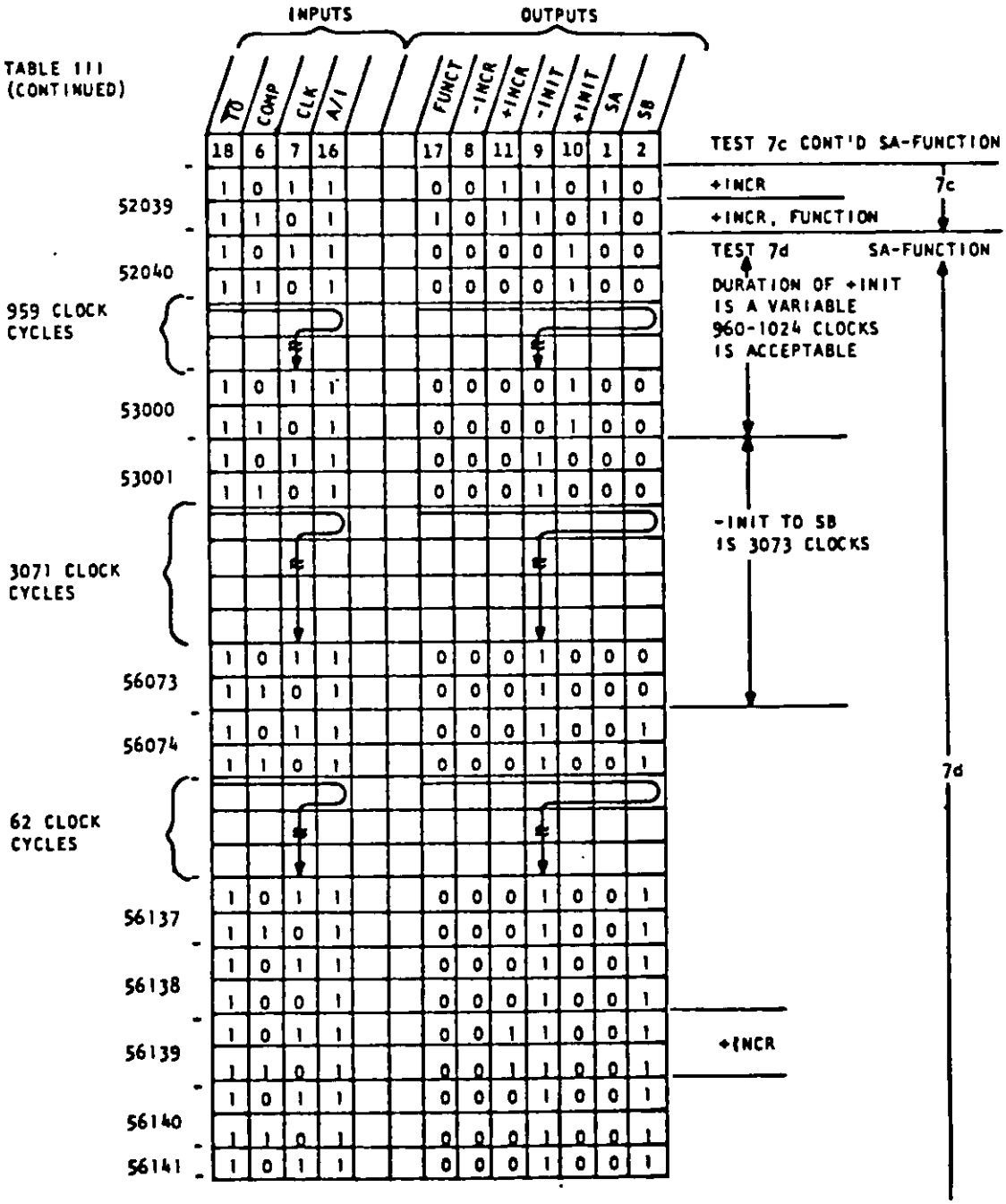


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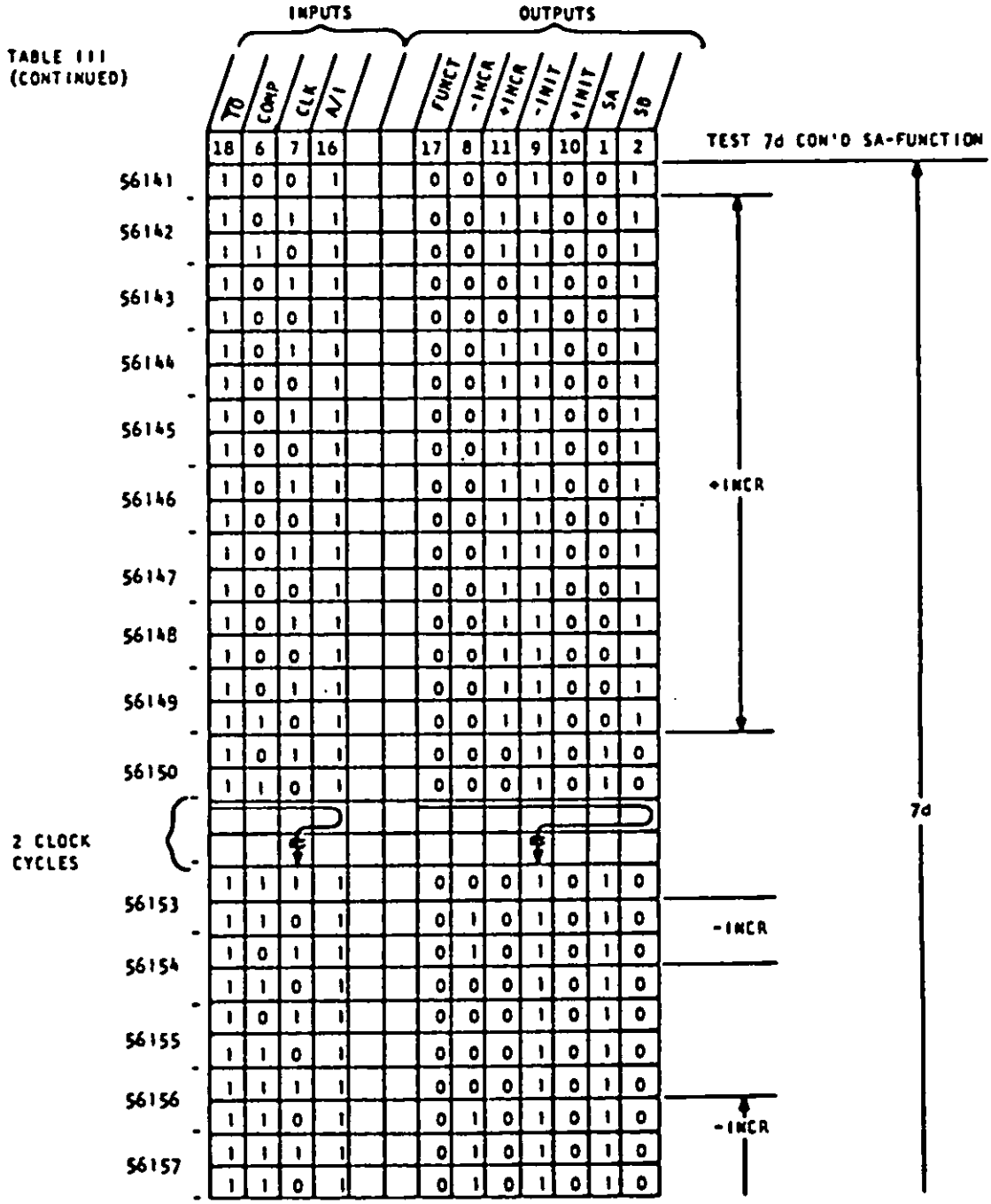


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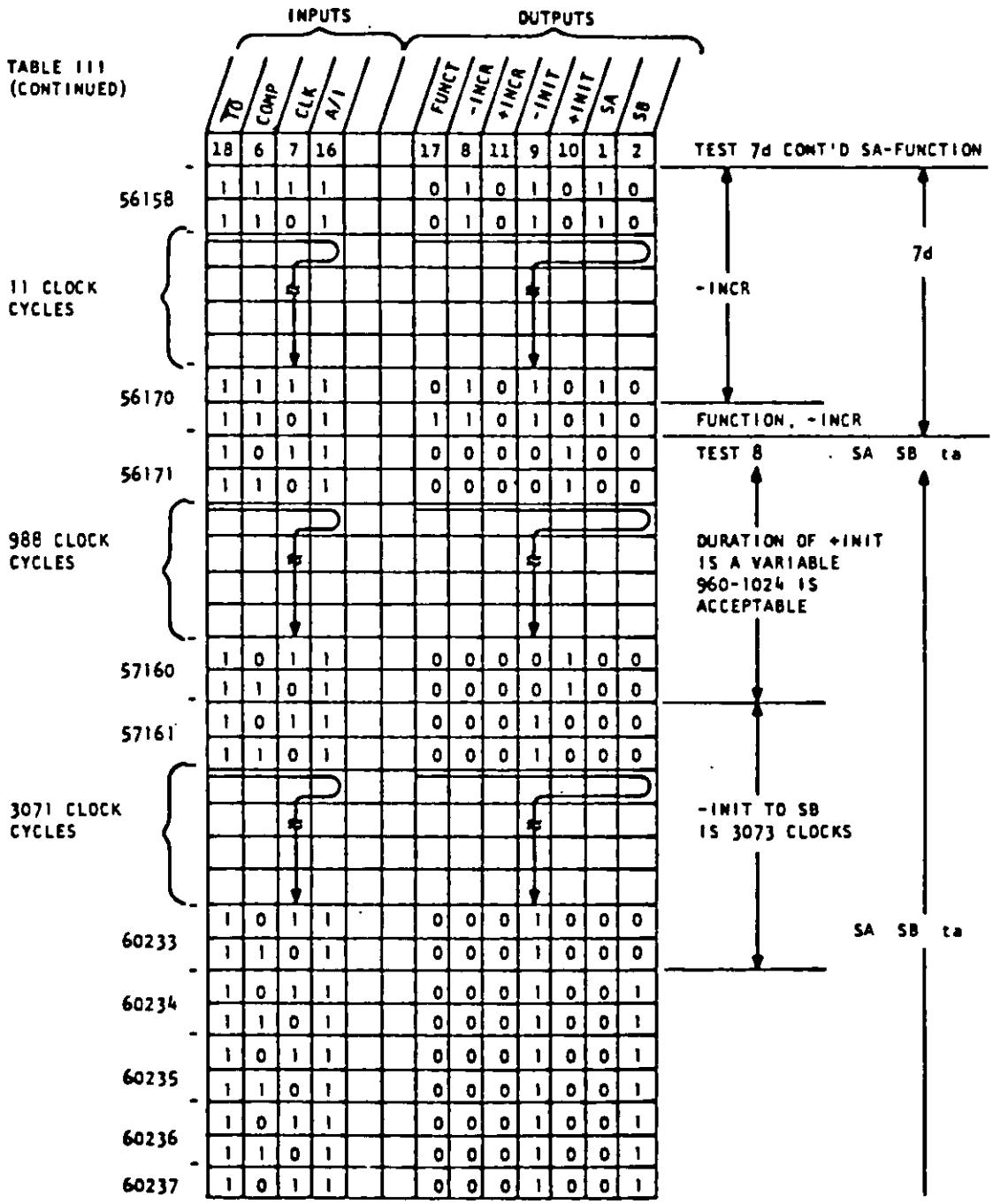


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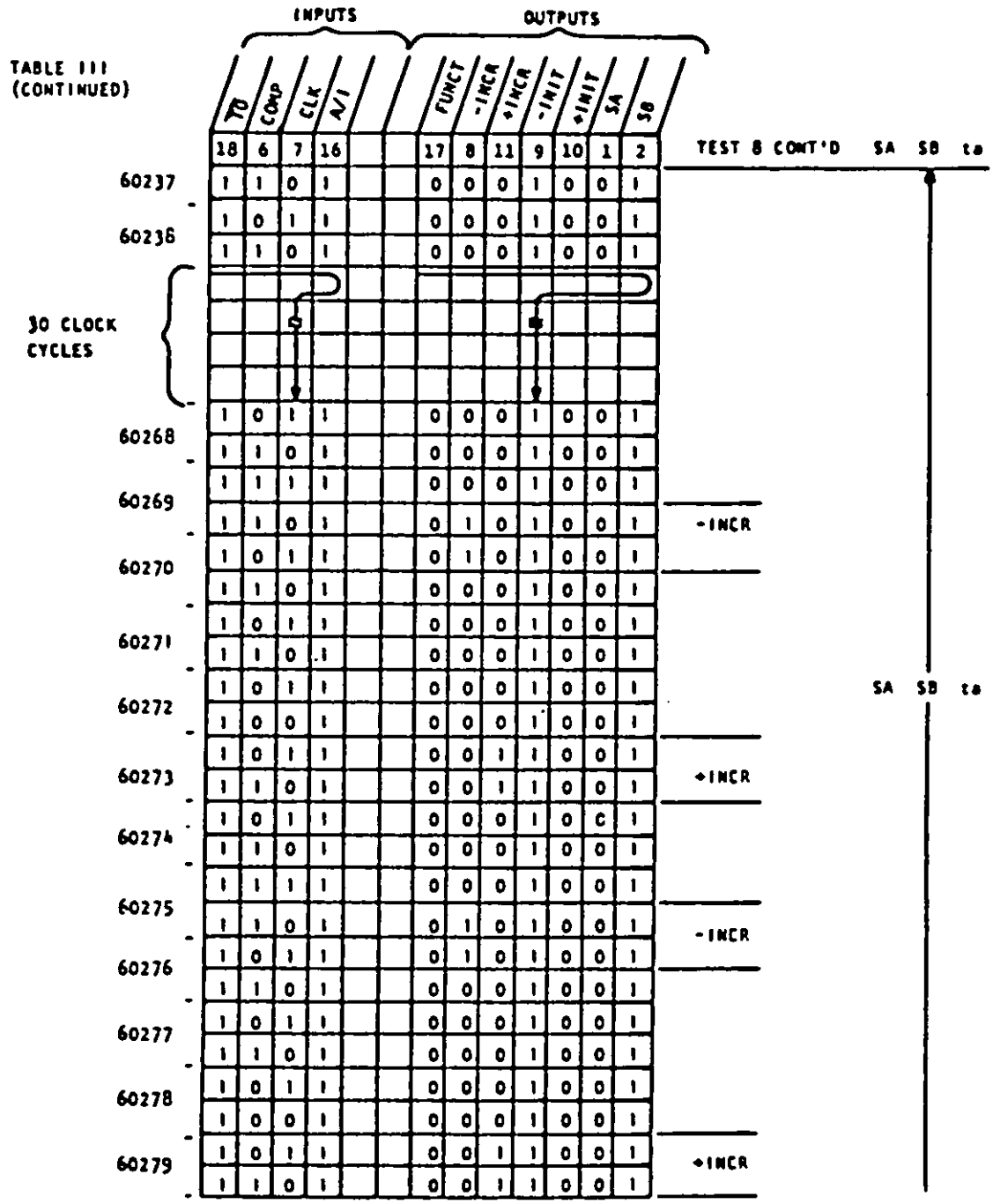


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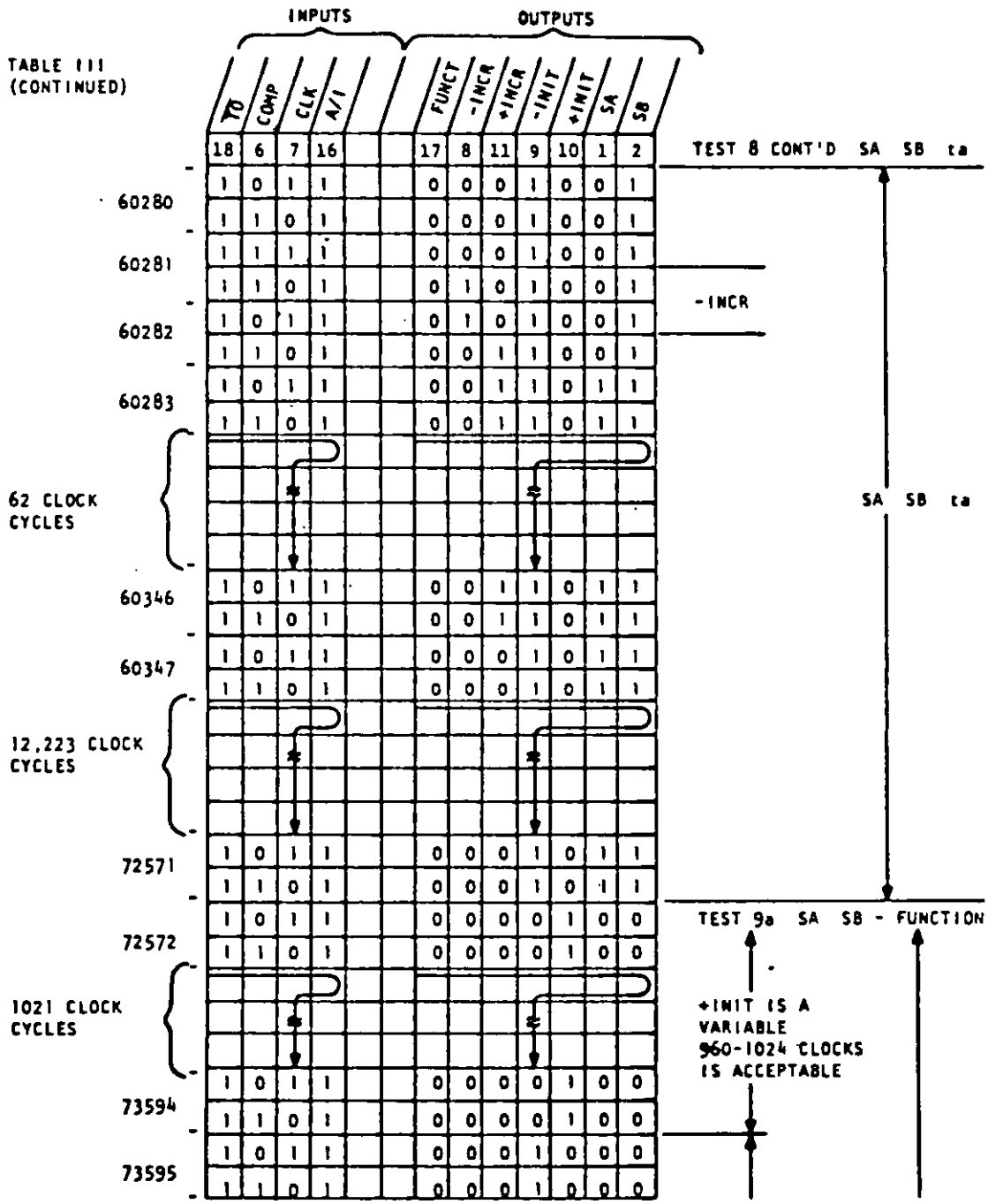


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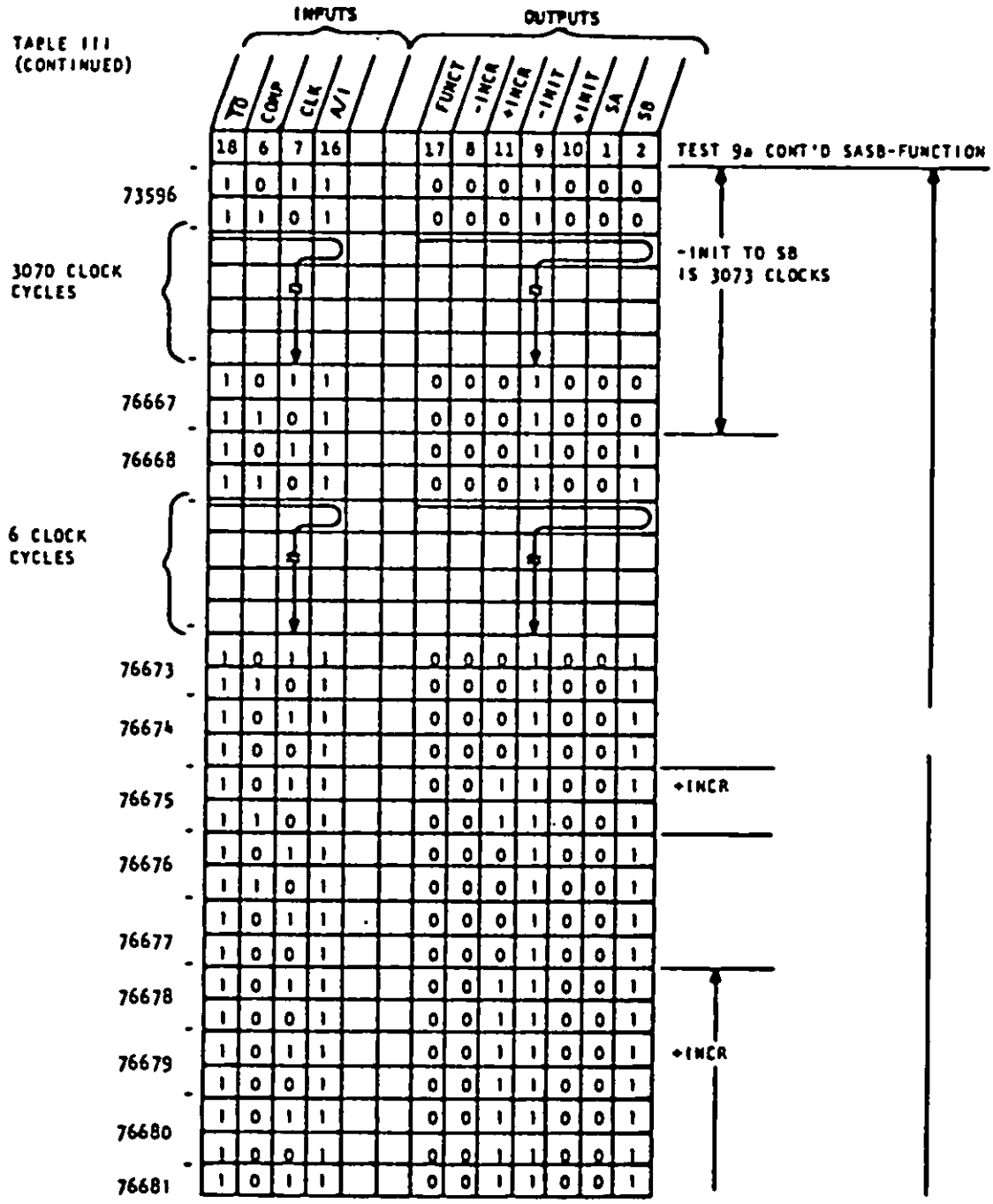


TABLE III. (Cont'd)

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TABLE III
(CONTINUED)

	INPUTS					OUTPUTS							
	TO	COMP	CLK	A/I		FUNCT	-INCR	+INCR	-INIT	+INIT	SA	SB	
	18	6	7	16		17	8	11	9	10	1	2	TEST 9a CONT'D SASB-FUNCTION
76681	1	0	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	+INCR
76682	1	1	0	1		0	0	1	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
76683	1	1	0	1		0	0	0	1	0	0	1	
	1	1	1	1		0	0	0	1	0	0	1	
76684	1	1	0	1		0	1	0	1	0	0	1	-INCR
	1	0	1	1		0	1	0	1	0	0	1	
76685	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
76686	1	1	0	1		0	0	0	1	0	0	1	
	1	1	1	1		0	0	0	1	0	0	1	
76687	1	1	0	1		0	1	0	1	0	0	1	
	1	1	1	1		0	1	0	1	0	0	1	
76688	1	1	0	1		0	1	0	1	0	0	1	-INCR
	1	0	1	1		0	1	0	1	0	0	1	
76689	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
76690	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
76691	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
76692	1	1	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	0	1	0	0	1	
76693	1	1	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	0	1	0	0	1	
76694	1	0	0	1		0	0	0	1	0	0	1	
	1	0	1	1		0	0	1	1	0	0	1	
76695	1	0	0	1		0	0	1	1	0	0	1	+INCR
	1	0	1	1		0	0	1	1	0	0	1	
76696	1	0	0	1		0	0	1	1	0	0	1	

9a

TABLE III. (Cont'd)

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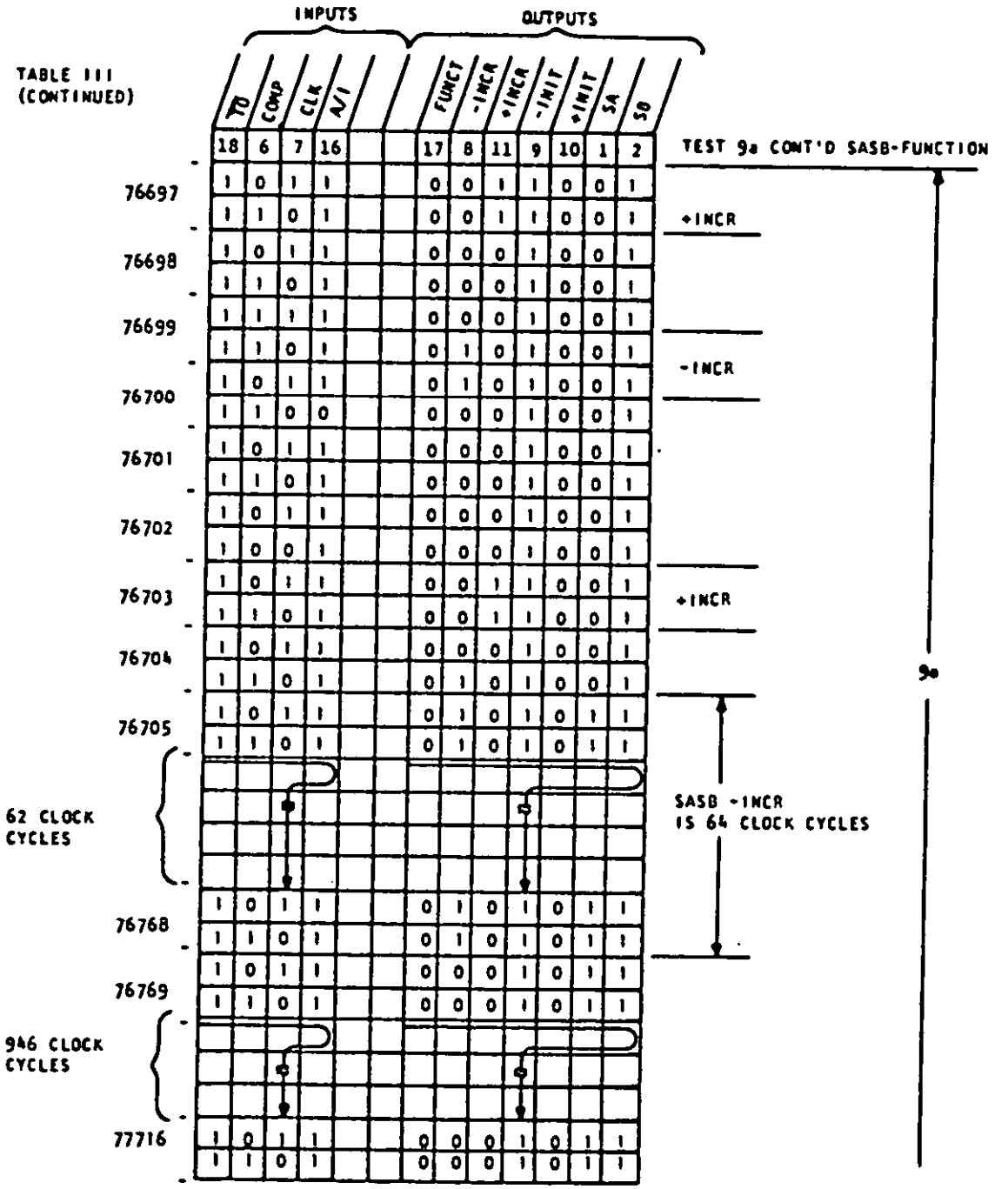


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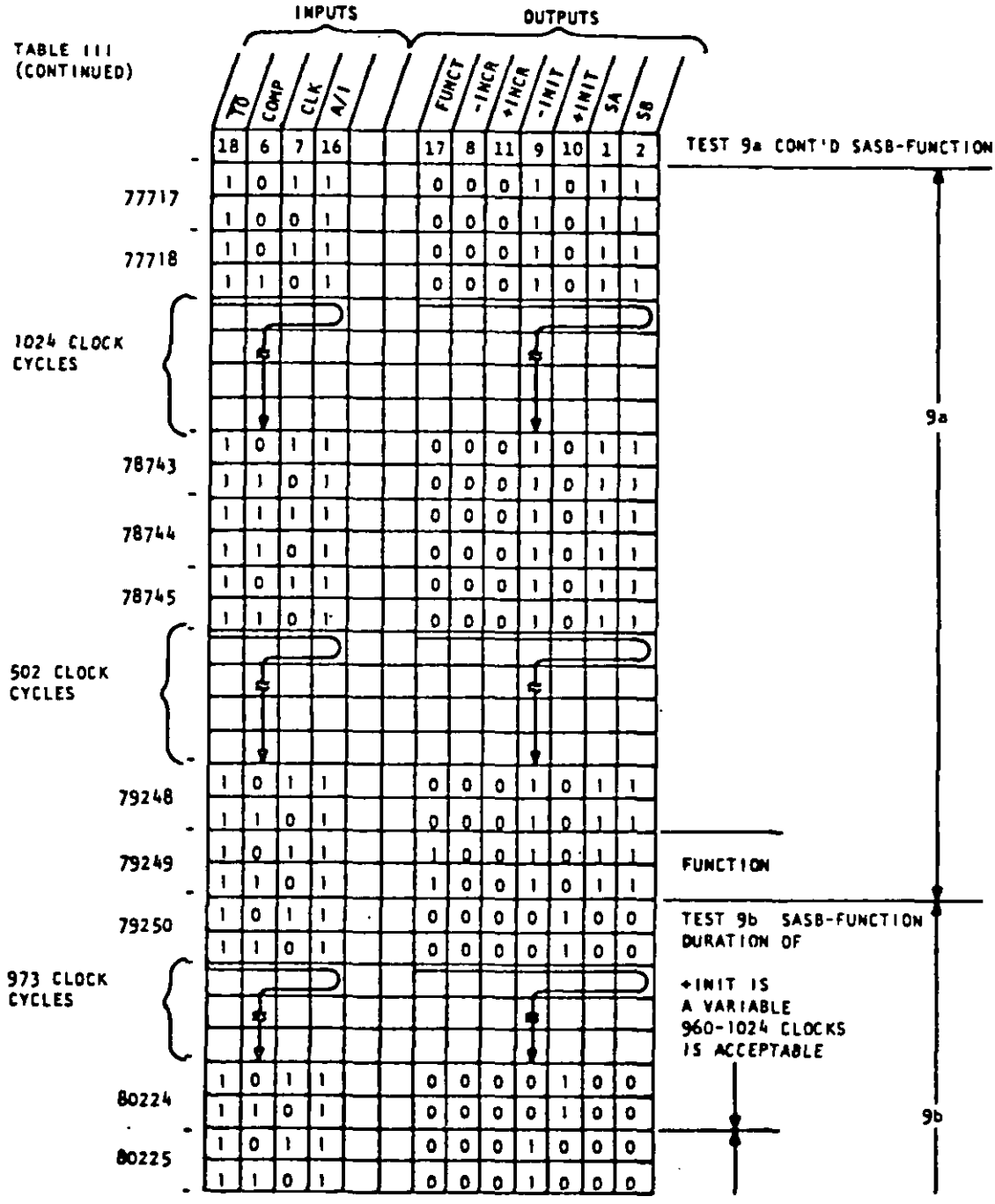


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TABLE III
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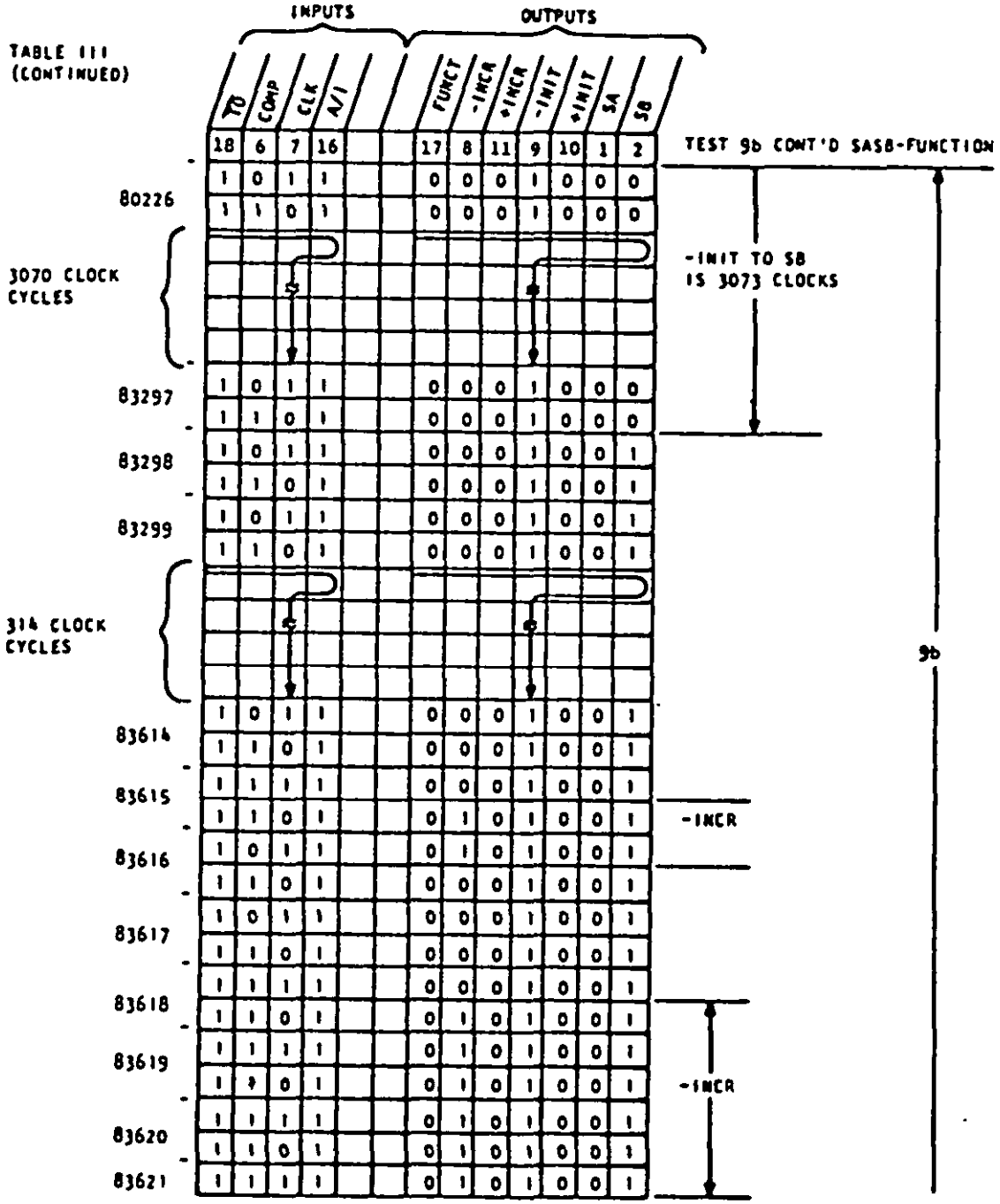


TABLE III. (Cont'd)

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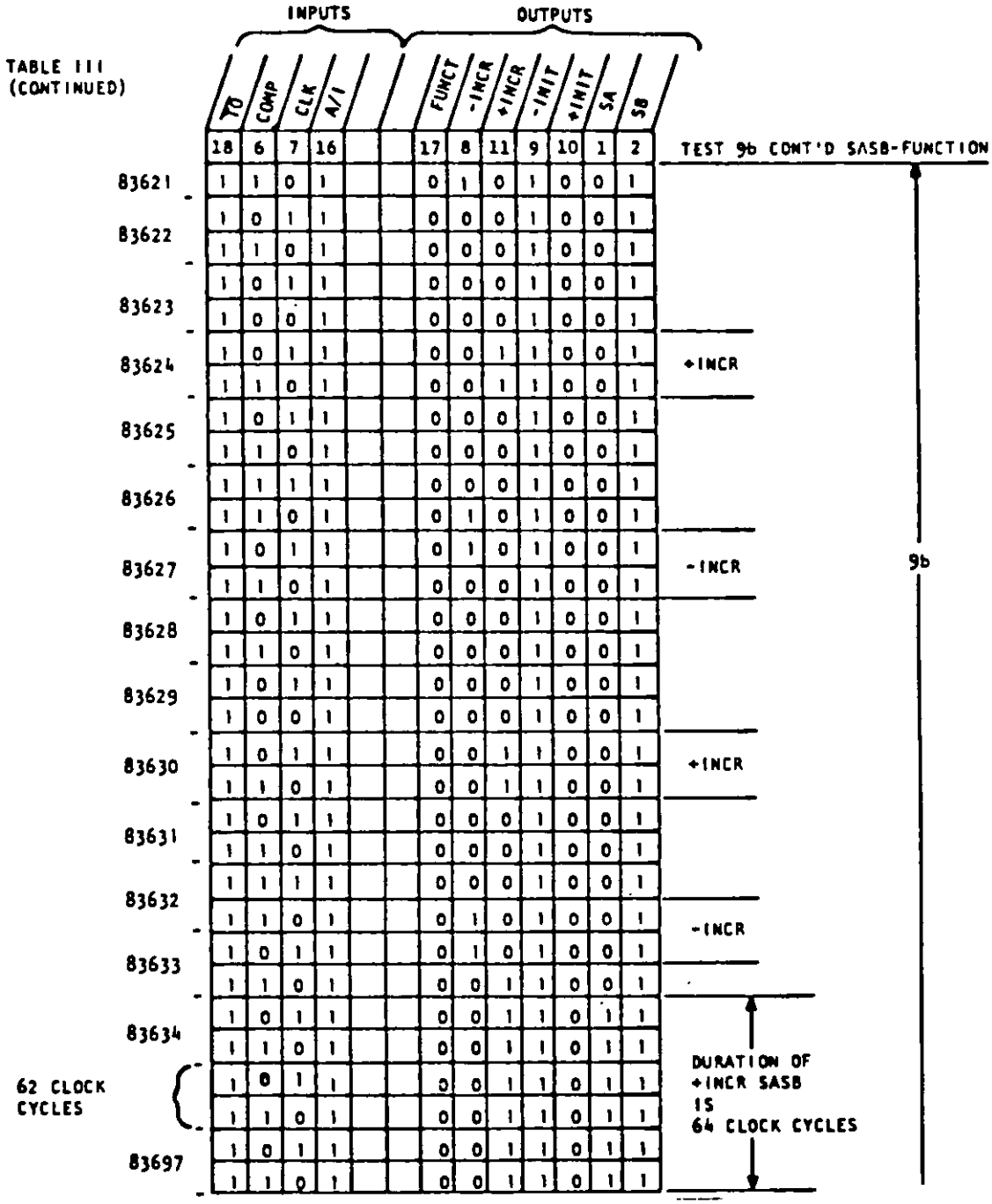


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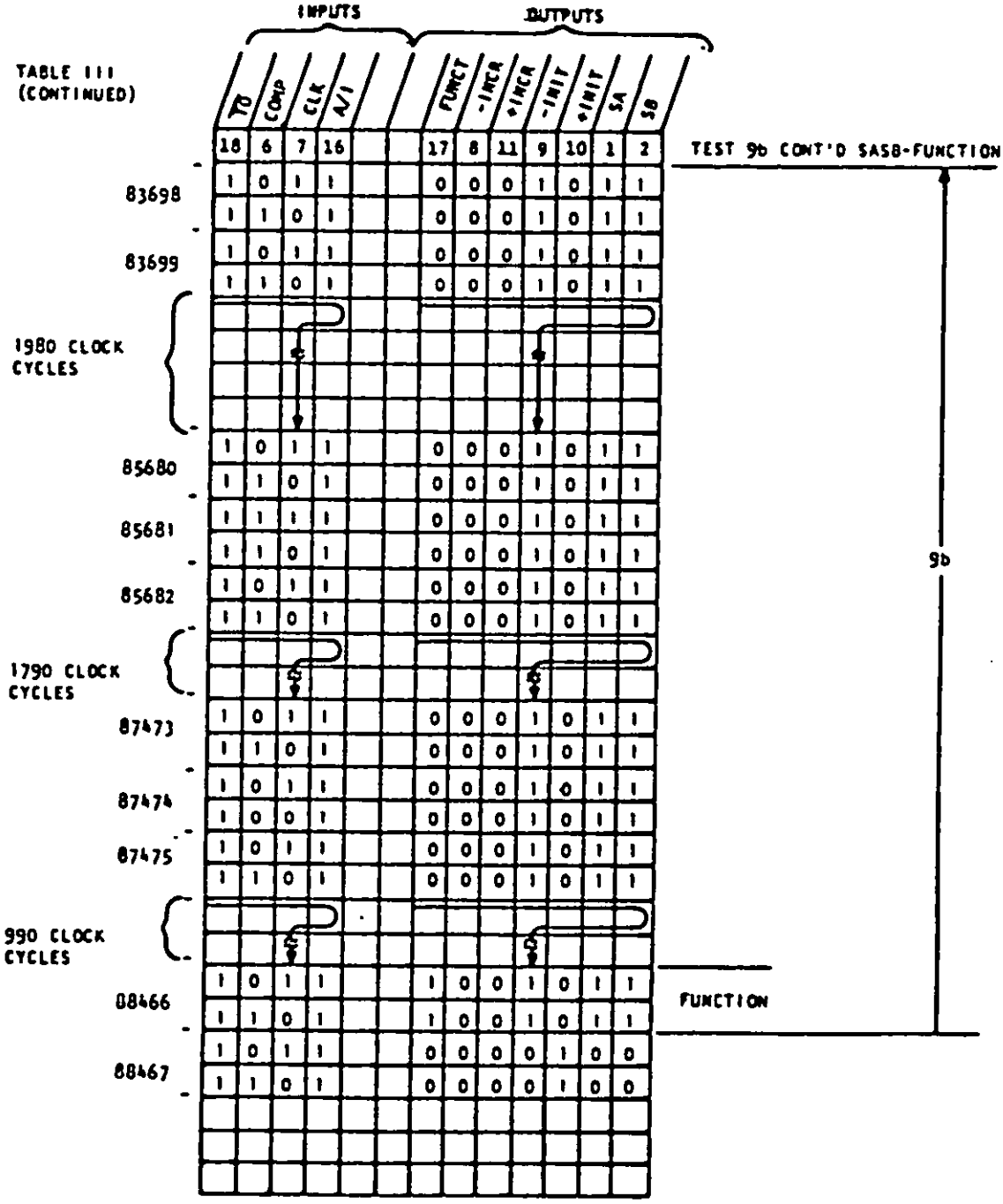


TABLE II. (Cont'd)

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3.5 Performance.

3.5.1 High temperature storage. I.C.'s shall be subjected to testing in accordance with MIL-STD-883, Method 1008, Test Condition C, and shall meet the final electrical requirements as specified in paragraph 3.4.2. There shall be no evidence of visual damage due to temperature exposure.

3.5.2 Temperature cycling. I.C.'s when tested as specified in MIL-STD-883, Method 1010, Test Condition C, shall meet the final electrical requirements listed in the paragraph 3.5.15.

3.5.3 Solderability. External leads, including tapered portion, must be electro tin plated or soldered dipped per MIL-P-38510, paragraph 3.5.6.2 (a) or (b) and if electro tin plated, the minimum plating must be 150 microinches. Parts must be solderable per MIL-STD-883, Method 2003.

3.5.4 Seal.

3.5.4.1 Fluorocarbon gross leak. I.C.'s shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition C.

3.5.4.2 Tracer gas (HE) fine leak. I.C.'s shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition A or B.

3.5.5 Thermal shock. I.C.'s, when subjected to the shock test specified in MIL-STD-883, Method 1011, Test Condition C, 15 cycles shall meet the final electrical requirements as specified in para 3.5.15 after exposure.

3.5.6 Constant acceleration. I.C.'s, when subjected to the shock test specified in MIL-STD-883, Method 2002, Test Condition E, Y_1 axis only, shall meet final electrical requirements as specified in paragraph 3.5.15.

3.5.7 Bond strength. Wires from the circuit die to the package header shall be aluminum with minimum bond strength equal to 3.0 gram pre-seal force. Bond strength shall be sufficient to meet the Class B requirements of MIL-STD-883, 2011.

3.5.8 Resistance to solvent. The marking of I.C.'s shall remain legible when subjected to resistance to solvent testing of MIL-STD-883, Method 2015, except paragraph 2.1.d.

3.5.9 Physical dimensions. I.C.'s shall comply with requirements specified in MIL-STD-883, Method 2016.

3.5.10 Internal visual. I.C.'s shall comply with requirements of MIL-STD-883, Method 2010, Condition B.

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3.5.11 External visual. The finished component shall meet the requirements of MIL-STD-883, Method 2009, there shall be no evidence of nonconformance with the detailed drawings, the absence of any required feature, damage, corrosion or contamination.

3.5.12 Lid torque. I.C.'s, when tested as specified in MIL-STD-883, Method 2024, shall exhibit no evidence of breakage, loosening, separation, or relative motion between the lid and the base when examined at a magnification between 10X and 20X.

3.5.13 First article. This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.5.14 Workmanship. All I.C.'s shall be manufactured, processed and tested in a careful and workmanship manner in accordance with good engineering practice and in accordance with the requirements of the specification.

3.5.15 Pre-seal bake. Eutectic attached substrate and sealing glass shall be baked at 125°C in a dry nitrogen atmosphere for a specified interval (16 hours minimum for eutectic attached substrate and 2 hours minimum for sealing glass) immediately prior to sealing.

3.5.16 Screening. Prior to final inspection, the finished component shall be screened in accordance with paragraph 4.6. Components subjected to the screening procedure shall be capable of functioning and meeting the requirements of 3.4.

3.5.17 Mechanical shock. The I.C.'s, when loaded in a 155MM test vehicle projectile, shall be capable of surviving the mechanical shock requirement as specified in drawing 9317646.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2.

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4.3.2 Inspection to be performed. The first article parts submitted in accordance with Paragraph 4.3.1 shall be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection of Table IV. All defects are classified as Major unless otherwise specified.

4.3.3 Inspection conditions. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

a.	Temperature	25 ± 5°C
b.	Humidity	10 to 75% RH
c.	Pressure	29 ± 3 inches of Hg
d.	Voltage and current measure	± 2%

4.3.4 Rejection. See MIL-A-48078. Resubmission for the first article tests shall be in accordance with the appropriate contract clause.

4.3.5 Re-qualification. I.C.'s shall be subject to another complete qualification if any changes are made which require an interfix number change (see paragraph 5.3).

4.4 Quality conformance inspection. The quality conformance inspection listed in paragraph Table V shall be performed on each inspection lot submitted for acceptance.

4.4.1 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Rejection. If any unit under test fails to comply with any of the applicable requirements, the unit shall be rejected. Lots shall not be resubmitted for these tests without prior approval of the procuring activity. Acceptance of product shall be discontinued pending investigation of the failure and approval by the design activity of corrective action proposed by the contractor.

4.5 Electrical tests.

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TABLE IV. QUALIFICATION TEST

SYMBOL	EXAMINATION OR TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED
	GROUP 1-A							
	In Process Screening		3.5.16	4.6				100%
	GROUP 1-B			MIL-STD-883				
	Physical Dimension		3.5.9	2016				MIL-STD-883
	Marking/Resistance to Solvent		3.5.8	2015 except para 2.1.d				Method 5005 Class B
	External Visual		3.5.11	2009				Sample Plan
	Internal Visual		3.5.10	2010				
	Bond Strength		3.5.7	2011				
	Lid Torque		3.5.12	2024				
	Thermal Shock		3.5.5	1011				
	Solderability		3.5.3	1010				
	GROUP II							LTPD
	(End Point Electrical Tests)		3.4.2					3

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TABLE V. QUALITY CONFORMANCE TEST

SYMBOL	EXAMINATION TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED
	GROUP 1 In Process Screening		3.5.16	4.6				100%
	Physical Dimensions		3.5.9	2016				MIL-STD-883
	Marking/Resistance to Solvent		3.5.8	2015 except para 2.1.d				Method 5005
	External Visual		3.5.11	2009				Class B
	Bond Strength		3.5.7	2011				Sample Plan
	Lid Torque		3.5.12	2024				
	Thermal Shock		3.5.5	1011				
	Solderability		3.5.3	1010				
	GROUP II (End point Electrical tests)		3.4.2					LTPD 5

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4.5.1 Test method. The test methods which follow are intended to convey the idea of what is to be tested and provide a simple bench method of doing it. It is recognized that automatic (computerized) tests will differ as to implementation of the test.

4.5.2 Leakage current. The microcircuit shall be connected as shown in Figure 3. The supply voltage shall be applied between terminals 5 (V_B) and 14 (ground). A minimum of 50 milliseconds shall elapse before measuring the leakage current. Leakage shall be as specified in TABLE II.

4.6 Screening. Screening shall be in accordance with MIL-STD-883, method 5004 with the exception of the following:

Screen	Method
Burn-In Test	1015 at 125°C for 160 hours
Post Burn-In	1015 at 125°C ALL ELECTRICAL Parameter per TABLE II.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Circuit shall be packaged with a conductive material so that damage from static charge or stray voltage is prevented.

5.2 Marking. Each circuit shall be marked with the last three digits of the part number and revision letter. (Example: 165A) The circuit shall also be marked with a lot number traceable to the vendor's manufacturing records. The vendor may add additional markings for his use provided the additional markings do not interfere with the required markings.

5.3 Inspection lot formation (Lot Identification). To meet the requirements for material traceability required by contract, each integrated circuit lot will contain only one interfix number.

The lot identification/date number shall be preceded by a three-digit interfix number. The interfix number shall be an integral part of the lot identification number, and must be used to identify those lots produced homogenously in one continuous process. Any of the conditions described below shall be cause for changing the interfix number.

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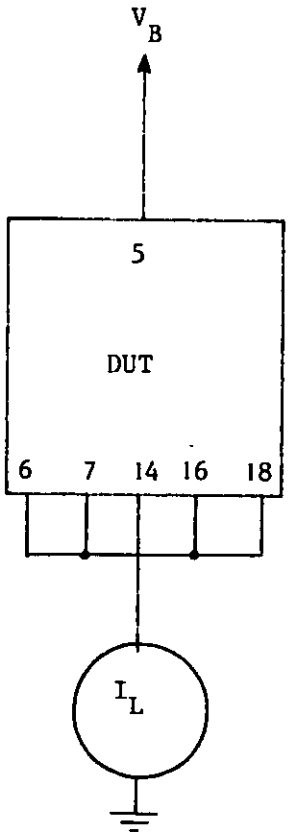


Figure 3. Leakage Current Test

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5.3.1 Change in manufacturing process. Changes representative of those which are subject to this requirement are any basic design changes in the manufacturing process such as:

a. Die structure topography changes (e.g., double-diffused, epitaxial, isolation).

b. Mask changes or redesign which alter die size.

c. Change in passivation or classification material thickness or technique (including addition or deletion of passivation).

d. Change in die substrate or element attach material, method or location.

e. Changes in wire dimensions.

f. Package or lid structure and material changes or changes to the internal cavity geometry.

5.3.2 Change in manufacturing location. If the active line moves from one geographical location to another, the interfix number will be changes.

5.3.3 Interruption in manufacturing process over 90 days. A complete line shutdown of more than 90 days duration will be cause for a change of the interfix number.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (PA)

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to Commander: USAMCCOM, ATTN: AMSMC-QAT-I(D), Dover, NJ 07801-5001. Request that the letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.4 Distribution of ammunition data cards, inspection and test results. Distribution of ammunition data cards, inspection, and test results shall include Commander: AMCCOM, Dover, NJ 07801-5001, ATTN: AMSMC-QAT-M(D).

6.5 Subject term (key word) listing.

Constant acceleration
Leakage Current test
Shock, Mechanical
Stress Screening
Power Supply requirement
Test Pattern
Requalification
High temperature storage

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A301)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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